CLASS BASED MODELLING

Document structure

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Modelling rule. The early stages remain intentionally broad. A noun that passes GC is only a candidate. Similar GC candidates are merged before SC. A merged candidate becomes a selected class when it satisfies at least three SC points or the Essential Requirement criterion. Finalization then confirms inheritance, responsibility placement and collaboration.

1. List of Nouns in Solution Space

The first step is to extract nouns and noun phrases from the complete usage scenario without deciding whether they will become classes. Roles, records, statuses, events, values, interfaces, reports, places and management concepts are all retained at this stage.

Table 1: Complete List of Solution-Space Nouns

No.	Noun
1	Ramna Yoga Society Management System
2	System
3	Digital support
4	Membership
5	Attendance
6	Payment
7	Finance
8	Event
9	Notice
10	Yoga video
11	Equipment
12	Committee activity
13	Yoga knowledge assistance
14	Regular yoga session
15	Physical yoga session
16	Registered member
17	Authorized committee member
18	System role
19	General participant
20	General participant interaction
21	Public information
22	Yoga activity
23	Regular yoga-session schedule
24	Schedule
25	Time
26	Location
27	Public notice
28	Upcoming event
29	Public yoga video
30	General safety guidance
31	Account
32	Membership application
33	Membership applicant
34	Membership applicant interaction
35	Membership registration
36	Applicant
37	Digital membership form
38	Required personal information
39	Photograph
40	Application
41	Treasurer

No.	Noun
42	Executive Committee
43	Information
44	Verification
45	Formal approval
46	Registration payment
47	Membership creation
48	BDT 500 registration fee
49	Supported payment facility
50	Accepted payment method
51	Payment received
52	Membership record
53	Unique membership number
54	QR code
55	Physical membership card
56	Registered member interaction
57	Member dashboard
58	Personal profile information
59	Membership status
60	Attendance history
61	Payment history
62	Donation history
63	Upcoming session
64	Available yoga video
65	Payments and donations
66	Member
67	Monthly payment
68	Yearly payment
69	Donation
70	Applicable event payment
71	Successful payment
72	Corresponding member
73	Digital confirmation
74	Receipt
75	Event participation
76	Event registration
77	Advance registration
78	QR-based attendance
79	Executive Committee interaction
80	Attendance management
81	Selected yoga session
82	Selected event
83	Registered member's QR card
84	Member identity
85	Member status
86	Attendance record
87	Duplicate attendance
88	Unavailable QR card

No.	Noun
89	Manual attendance
90	Member's membership number
91	Attendance summary
92	High attendance
93	Session management
94	Yoga-session date
95	Starting time
96	Instructor
97	Session status
98	Normal session time
99	One hour
100	Fajr prayer
101	Committee administration
102	System permission
103	Official position
104	Committee membership
105	Administrative permission
106	Previous committee record
107	Treasurer interaction
108	Membership and payment management
109	Membership applications
110	Applicant information
111	Membership records
112	Payments
113	Donations
114	Online-payment information
115	Overdue membership
116	Financial management
117	Income
118	Expense
119	Financial total
120	Balance information
121	Monthly financial report
122	Yearly financial report
123	President
124	Society responsibility structure
125	Financial record
126	President and Secretary interaction
127	Secretary
128	Organizational supervision
129	Membership summary
130	Financial summary
131	Event summary
132	Notice summary
133	Organizational summary
134	Important committee activity
135	Financial-recording responsibility

No.	Noun
136	Report review
137	Financial report
138	Attendance information
139	Membership information
140	Organizational report
141	Yoga Instructor interaction
142	Yoga content
143	Safety guidance
144	Yoga Instructor
145	Yoga-pose information
146	Instruction
147	Difficulty level
148	Possible risk
149	Age-related precaution
150	Instructor-approved guidance
151	Pose
152	Medical suitability
153	Particular person
154	Yoga video management
155	Approved yoga video
156	Video title
157	Video instructor
158	Video category
159	Video difficulty level
160	Safety information
161	Public video visibility
162	Members-only video visibility
163	User
164	Vajra
165	AI Yoga Knowledge Assistant
166	Vajra interaction
167	Member question
168	General visitor
169	Question
170	Yoga pose
171	Pose instruction
172	Precaution
173	Instructor information
174	Yoga information
175	Approved instructor-provided content
176	Short conversational answer
177	Relevant video
178	Learning material
179	Safety-related question
180	Authorized yoga instructor
181	Medical condition
182	Specific person

No.	Noun
183	Publicity Secretary interaction
184	Publicity Secretary
185	Notice and communication management
186	Official notice
187	Regular activity
188	Session change
189	Postponement
190	Cancellation
191	Announcement
192	User group
193	Previous notice
194	Media management
195	Activity post
196	Description
197	Activity photograph
198	Activity video
199	Activity date
200	Participation information
201	Media
202	Yoga-session media
203	Event media
204	Public media visibility
205	Members-only media visibility
206	Cultural Secretary interaction
207	Cultural Secretary
208	Event management
209	Cultural event
210	Special event
211	Event date
212	Event time
213	Event location
214	Event description
215	Event instruction
216	Registration information
217	Participant information
218	Authorized user
219	Financial matter
220	Event publicity
221	Field and Equipment Secretary interaction
222	Field and Equipment Secretary
223	Equipment management
224	Equipment information
225	Quantity
226	Condition
227	Purchase information
228	Storage location
229	Equipment assignment

No.	Noun
230	Regular-session equipment assignment
231	Event equipment assignment
232	Helping-Hand Management
233	Task
234	Helping hand
235	Yoga area
236	Equipment arrangement
237	Event support
238	Equipment return
239	Responsible committee member
240	Task progress
241	Helping-hand payment information
242	Automatic system interaction
243	Member engagement
244	QR attendance
245	Motivational message
246	Attendance streak
247	Attendance milestone
248	Birthday
249	Membership anniversary
250	Birthday greeting
251	Membership-anniversary greeting
252	Special-day appearance
253	System appearance
254	Selected occasion
255	International Mother Language Day
256	Independence Day
257	International Yoga Day
258	Victory Day
259	Visual change
260	Regular system operation

2. General Classifications (GC)

The extracted nouns are examined through seven General Classifications. These categories are used as non-exclusive analytical lenses: one noun may satisfy several classifications. A noun passes GC when it satisfies at least three classifications, or when it satisfies External Entity by itself.

Table 2: Seven General Classifications

No.	General Classification	Description
1	External entities	People, organizations, devices or external services that exchange information with the system.
2	Things	Physical or informational objects that the system stores, manages or manipulates.
3	Events	Occurrences, transactions or lifecycle changes that the system must recognize or record.
4	Roles	Capacities performed by people or software actors.
5	Organizational units	Groups or functional management units with coordinated responsibilities.
6	Places	Physical or logical locations relevant to system interaction.
7	Structures	Composite records, aggregates, hierarchies or organized information structures.

Table 3: Nouns Passing General Classification

No.	Noun	Accepted GC	Count	Result
1	General participant	1, 2, 4	3	Pass
2	Registered member	1, 2, 4	3	Pass
3	Membership applicant	1, 2, 4	3	Pass
4	Applicant	1, 2, 4	3	Pass
5	Member	1, 2, 4	3	Pass
6	Instructor	1, 2, 4	3	Pass
7	Yoga Instructor	1, 2, 4	3	Pass
8	User	1, 2, 4	3	Pass
9	General visitor	1, 2, 4	3	Pass
10	Helping hand	1, 2, 4	3	Pass
11	Authorized user	1, 2, 4	3	Pass
12	Authorized committee member	1, 2, 4, 5	4	Pass
13	Treasurer	1, 2, 4, 5	4	Pass
14	Authorized Executive Committee member	1, 2, 4, 5	4	Pass
15	President	1, 2, 4, 5	4	Pass
16	Secretary	1, 2, 4, 5	4	Pass
17	Publicity Secretary	1, 2, 4, 5	4	Pass
18	Cultural Secretary	1, 2, 4, 5	4	Pass
19	Field and Equipment Secretary	1, 2, 4, 5	4	Pass
20	Responsible committee member	1, 2, 4, 5	4	Pass
21	Executive Committee	1, 5, 7	3	Pass
22	Supported payment facility	1, 2, 7	3	Pass
23	Membership	2, 3, 7	3	Pass
24	Attendance	2, 3, 7	3	Pass
25	Payment	2, 3, 7	3	Pass
26	Event	2, 3, 7	3	Pass
27	Notice	2, 3, 7	3	Pass
28	Committee activity	2, 3, 7	3	Pass
29	Yoga knowledge assistance	2, 3, 7	3	Pass
30	Regular yoga session	2, 3, 7	3	Pass
31	Physical yoga session	2, 3, 7	3	Pass
32	Membership application	2, 3, 7	3	Pass
33	Membership registration	2, 3, 7	3	Pass

No.	Noun	Accepted GC	Count	Result
34	Application	2, 3, 7	3	Pass
35	Registration payment	2, 3, 7	3	Pass
36	Membership creation	2, 3, 7	3	Pass
37	Donation	2, 3, 7	3	Pass
38	Event participation	2, 3, 7	3	Pass
39	Event registration	2, 3, 7	3	Pass
40	QR-based attendance	2, 3, 7	3	Pass
41	Attendance record	2, 3, 7	3	Pass
42	Income	2, 3, 7	3	Pass
43	Expense	2, 3, 7	3	Pass
44	Yoga content	2, 3, 7	3	Pass
45	Activity post	2, 3, 7	3	Pass
46	Cultural event	2, 3, 7	3	Pass
47	Special event	2, 3, 7	3	Pass
48	Equipment assignment	2, 3, 7	3	Pass
49	Task	2, 3, 7	3	Pass
50	Equipment arrangement	2, 3, 7	3	Pass
51	Equipment return	2, 3, 7	3	Pass
52	Automatic system interaction	2, 3, 7	3	Pass
53	Member engagement	2, 3, 7	3	Pass
54	Attendance streak	2, 3, 7	3	Pass
55	Attendance milestone	2, 3, 7	3	Pass
56	Membership anniversary	2, 3, 7	3	Pass
57	Special-day appearance	2, 3, 7	3	Pass
58	System appearance	2, 3, 7	3	Pass
59	Attendance management	2, 3, 5, 7	4	Pass
60	Session management	2, 3, 5, 7	4	Pass
61	Committee administration	2, 3, 5, 7	4	Pass
62	Membership and payment management	2, 3, 5, 7	4	Pass
63	Financial management	2, 3, 5, 7	4	Pass
64	Organizational supervision	2, 3, 5, 7	4	Pass
65	Yoga video management	2, 3, 5, 7	4	Pass
66	Notice and communication management	2, 3, 5, 7	4	Pass
67	Media management	2, 3, 5, 7	4	Pass
68	Event management	2, 3, 5, 7	4	Pass
69	Equipment management	2, 3, 5, 7	4	Pass
70	Helping-Hand Management	2, 3, 5, 7	4	Pass
71	Account	2, 4, 7	3	Pass
72	Vajra	2, 4, 7	3	Pass
73	AI Yoga Knowledge Assistant	2, 4, 7	3	Pass
74	Member dashboard	2, 6, 7	3	Pass
75	Committee membership	2, 3, 4, 5, 7	5	Pass
76	Ramna Yoga Society Management System	2, 5, 7	3	Pass

GC result: 76 of the 260 extracted nouns pass. The remaining nouns remain useful as possible attributes, values, reports or method concepts, but they are not promoted to potential classes.

3. Selection Criteria (SC)

Before applying Selection Criteria, semantically equivalent or strongly overlapping GC-passed nouns are merged. This prevents the same domain concept from being scored repeatedly under different wording.

3.1 Merging and Renaming the 76 GC-Passed Nouns

Table 4: Merging GC-Passed Nouns into Potential Classes

No.	GC-passed nouns merged	Merged potential class	Count
1	Ramna Yoga Society Management System	SystemBoundary	1
2	General participant, General visitor	Visitor	2
3	User, Authorized user, Account	UserAccount	3
4	Membership applicant, Applicant	Applicant	2
5	Registered member, Member, Membership	Member	3
6	Membership application, Membership registration, Application, Membership creation	MembershipRegistration	4
7	Instructor, Yoga Instructor	YogaInstructor	2
8	Executive Committee, Committee administration, Organizational supervision	ExecutiveCommittee	3
9	Authorized committee member, Authorized Executive Committee member, Responsible committee member, Committee membership	ExecutiveCommitteeMember	4
10	President	President	1
11	Secretary	Secretary	1
12	Treasurer	Treasurer	1
13	Publicity Secretary	PublicitySecretary	1
14	Cultural Secretary	CulturalSecretary	1
15	Field and Equipment Secretary	FieldEquipmentSecretary	1
16	Committee activity	Activity	1
17	Regular yoga session, Physical yoga session, Session management	YogaSession	3
18	Event, Cultural event, Special event, Event management	Event	4
19	Event participation, Event registration	EventParticipation	2
20	Attendance, QR-based attendance, Attendance record, Attendance management	Attendance	4
21	Payment, Registration payment, Donation, Income, Expense	FinancialTransaction	5
22	Membership and payment management, Financial management	FinancialManagement	2
23	Supported payment facility	PaymentGateway	1
24	Notice, Activity post, Notice and communication management, Media management	MediaPost	4
25	Yoga content, Yoga video management	YogaContent	2
26	Yoga knowledge assistance, Vajra, AI Yoga Knowledge Assistant	VajraAssistant	3
27	Equipment assignment, Equipment arrangement, Equipment return, Equipment management	InventoryManagement	4
28	Task	Task	1
29	Helping hand, Helping-Hand Management	HelpingHand	2
30	Member dashboard	MemberDashboard	1
31	Automatic system interaction	SystemAutomation	1
32	Member engagement, Attendance streak, Attendance milestone, Membership anniversary	MemberEngagement	4
33	Special-day appearance, System appearance	SystemAppearance	2

3.2 Applying the Six Selection Criteria

The merged potential classes are evaluated using six criteria. A candidate passes when it satisfies at least three criteria, or when Essential Requirement is accepted. Essential Requirement means that the requirement needs an independent class, not merely that the associated feature must exist.

Table 5: Six Selection Criteria

No.	Selection Criterion	Description
1	Retained Information	Information about the candidate must be remembered by the system.

No.	Selection Criterion	Description
2	Needed Services	The candidate provides identifiable services required by the system.
3	Multiple Attributes	The candidate needs several meaningful attributes.
4	Common Attributes	Instances or related subclasses share attributes.
5	Common Operations	Instances or related subclasses share operations.
6	Essential Requirement	An independent class is essential to satisfy the requirement.

Table 6: Selection Criteria of Potential Classes

No.	Potential class	Accepted criteria	Points	Result
1	SystemBoundary	2	1	Fail
2	Visitor	2	1	Fail
3	UserAccount	1, 2, 3, 4, 5, 6	6	Pass
4	Applicant	2, 3	2	Fail
5	Member	1, 2, 3, 4, 5, 6	6	Pass
6	MembershipRegistration	2, 3	2	Fail
7	YogaInstructor	1, 2, 3, 4, 5, 6	6	Pass
8	ExecutiveCommittee	1, 3	2	Fail
9	ExecutiveCommitteeMember	1, 2, 3, 4, 5, 6	6	Pass
10	President	1, 2, 3, 4, 5, 6	6	Pass
11	Secretary	1, 2, 3, 4, 5, 6	6	Pass
12	Treasurer	1, 2, 3, 4, 5, 6	6	Pass
13	PublicitySecretary	1, 2, 3, 4, 5, 6	6	Pass
14	CulturalSecretary	1, 2, 3, 4, 5, 6	6	Pass
15	FieldEquipmentSecretary	1, 2, 3, 4, 5, 6	6	Pass
16	Activity	1, 2, 3, 4, 5, 6	6	Pass
17	YogaSession	1, 2, 3, 4, 5, 6	6	Pass
18	Event	1, 2, 3, 4, 5, 6	6	Pass
19	EventParticipation	1, 3	2	Fail
20	Attendance	1, 3	2	Fail
21	FinancialTransaction	3, 4	2	Fail
22	FinancialManagement	1, 2, 3, 4, 5, 6	6	Pass
23	PaymentGateway	1, 2, 3, 5, 6	5	Pass
24	MediaPost	1, 2, 3, 4, 5, 6	6	Pass
25	YogaContent	1, 2, 3, 4, 5, 6	6	Pass
26	VajraAssistant	1, 2, 3, 5, 6	5	Pass
27	InventoryManagement	1, 2, 3, 4, 5, 6	6	Pass
28	Task	1, 3	2	Fail
29	HelpingHand	1, 3	2	Fail
30	MemberDashboard	2, 3	2	Fail
31	SystemAutomation	2, 5	2	Fail
32	MemberEngagement	2, 3	2	Fail
33	SystemAppearance	2, 3	2	Fail

SC result: 19 potential classes pass. Fourteen candidates fail as independent classes. Their information and behaviour are not discarded; they are assigned to the nearest cohesive passing class during finalization.

4. Associate Noun and Verb Identification

For each selected class, associated nouns are identified as sources of attributes, while associated verbs are identified as sources of methods and responsibilities. The list remains domain-oriented and does not include framework, controller or database implementation operations.

Table 7: Associate Noun and Verb Identification

No.	Selected class	Associated nouns	Associated verbs
1	UserAccount	account identifier, username, password hash, account status, access permission, last login time	authenticate, log in, log out, change password, activate account, deactivate account, grant permission, revoke permission, update permission
2	Member	member identifier, full name, contact information, photograph, date of birth, application status, verification status, registration payment status, membership number, membership status, QR code, membership card, registration date, membership start date	register, submit application, update application, view status, verify information, request correction, record payment, activate membership, generate membership number, generate QR code, issue card, update profile, pay fee, donate, register for event, view histories, view activities, view content
3	YogaInstructor	instructor identifier, full name, contact information, expertise, yoga pose, instruction, difficulty, risk, precaution, video	create content, update content, add pose instruction, set difficulty, add risk, add precaution, attach demonstration video, submit content
4	ExecutiveCommitteeMember	committee member identifier, member, committee term, committee status, member application, attendance, summary, system permission	verify member information, request correction, record QR attendance, record manual attendance, generate summary, view directory, manage committee term, access administration
5	President	president role, committee term, financial report, attendance summary, committee activity, club operation	review report, review attendance, monitor committee, supervise operation
6	Secretary	secretary role, committee term, committee record, schedule, attendance summary, administrative activity	maintain record, coordinate schedule, review attendance, monitor administration
7	Treasurer	treasurer role, membership application, membership record, payment, donation, income, expense, balance, financial report	receive application, maintain membership record, record payment, record donation, record income, record expense, verify gateway payment, generate report
8	PublicitySecretary	publicity role, notice, announcement, activity post, media, attachment, audience, visibility	create post, publish notice, publish activity media, update post, archive post
9	CulturalSecretary	cultural role, event, registration, participant, event instruction, event publicity	create event, update event, open registration, manage participant, coordinate event
10	FieldEquipmentSecretary	field role, equipment, availability, assignment, return, operational task, assignee, progress	add equipment, update equipment, check availability, assign equipment, record return, create task, assign task, update progress
11	Activity	activity identifier, title, date, start time, location, status, organizer, participant, attendance entry	schedule, reschedule, cancel, add organizer, record attendance, summarize attendance, list participant
12	YogaSession	session, instructor, duration, regular-session status, schedule	assign instructor, open session, close session, publish schedule
13	Event	event type, description, instruction, registration status, registered member, participation information	open registration, register member, remove registration, close registration, publish participation information
14	FinancialManagement	membership payment, donation, income entry, expense entry, balance, report, payment history, donation history	record payment, record donation, record income, record expense, calculate balance, reconcile transaction, generate report, retrieve history
15	PaymentGateway	gateway name, supported method, availability, payment request, transaction reference, transaction status, confirmation	process payment, verify transaction, confirm payment, obtain status
16	MediaPost	post identifier, post type, title, body, attachment, visibility, audience, publish date, related activity	create, update, publish, set audience, attach media, archive, link activity
17	YogaContent	content identifier, content type, title, instruction, difficulty, risk, precaution, video reference, visibility, approval status, author	create, update, add instruction, set difficulty, add risk, add precaution, attach video, submit, approve, publish, search
18	VajraAssistant	approved content source, question, interpreted topic, answer, suggestion, safety disclaimer	interpret question, search approved content, generate answer, suggest related content, attach safety disclaimer
19	InventoryManagement	equipment catalogue, availability record, equipment assignment, operational task, task assignee, progress record, return record	add equipment, update equipment, remove equipment, check availability, assign equipment, record return, create task, assign task, update progress, close task

5. Attribute Selection

Attributes are selected from the associated nouns and refined to avoid duplication. Subclasses inherit shared attributes from their superclasses. Collection-like attributes represent contained records without creating unnecessary standalone classes at this high level.

Table 8: Attribute Selection of Classes

No.	Class	Selected attributes
1	UserAccount	accountId username passwordHash accountStatus accessPermissions lastLoginAt
2	Member	memberId fullName contactInformation photograph dateOfBirth applicationStatus verificationStatus registrationPaymentStatus membershipNumber membershipStatus qrCode cardInformation registrationDate membershipStartDate account
3	YogaInstructor	instructorId fullName contactInformation expertise account authoredContent
4	ExecutiveCommitteeMember	committeeMemberId member termStartDate termEndDate committeeStatus account assignedPermissions
5	President	inherits ExecutiveCommitteeMember attributes oversightNotes
6	Secretary	inherits ExecutiveCommitteeMember attributes committeeRecords coordinatedSchedules
7	Treasurer	inherits ExecutiveCommitteeMember attributes receivedApplications managedFinancialRecords
8	PublicitySecretary	inherits ExecutiveCommitteeMember attributes managedPosts
9	CulturalSecretary	inherits ExecutiveCommitteeMember attributes managedEvents
10	FieldEquipmentSecretary	inherits ExecutiveCommitteeMember attributes managedEquipment managedTasks
11	Activity	activityId title activityDate startTime location status organizers participants attendanceEntries

No.	Class	Selected attributes
12	YogaSession	inherits Activity attributes instructor duration regularSession
13	Event	inherits Activity attributes eventType description instructions registrationRequired registrationOpen registeredMembers participationInformation
14	FinancialManagement	membershipPayments donations incomeEntries expenseEntries currentBalance generatedReports
15	PaymentGateway	gatewayName supportedMethods availability transactionReferences lastGatewayResponse
16	MediaPost	postId postType title body attachments visibility targetAudience publishDate relatedActivity archiveStatus
17	YogaContent	contentId contentType title instructions difficultyLevel risks precautions videoReference visibility approvalStatus author createdAt updatedAt
18	VajraAssistant	approvedContentSource currentQuestion interpretedTopic generatedAnswer suggestions safetyDisclaimer
19	InventoryManagement	equipmentCatalogue availabilityRecords equipmentAssignments returnRecords operationalTasks taskAssignees progressRecords

6. Method Identification

Methods are derived from associated verbs and assigned to the class that owns the required information. The method list describes domain behaviour; validation and persistence helpers are intentionally omitted.

Table 9: Methods of Classes

No.	Class	Identified methods
1	UserAccount	authenticate() login() logout() changePassword() activateAccount() deactivateAccount() grantPermission() revokePermission() updatePermissions()
2	Member	static registerForMembership() submitApplication() updateApplication() viewApplicationStatus() markInformationVerified() requestApplicationCorrection() recordRegistrationPayment() activateMembership() generateMembershipNumber() generateQRCode() issueMembershipCard() updateProfile() payFee() donate() registerForEvent() viewPaymentHistory() viewDonationHistory() viewUpcomingActivities() viewYogaContent()
3	YogaInstructor	createYogaContent() updateYogaContent() addPoseInstructions() setDifficultyLevel() addRiskAndPrecaution() attachDemonstrationVideo() submitContentForApproval() viewAssignedSessions()
4	ExecutiveCommitteeMember	verifyMemberInformation() requestMemberCorrection() recordAttendanceByQR() recordAttendanceManually() generateAttendanceSummary() viewMemberDirectory() manageCommitteeTerm() createActivity() updateActivity()
5	President	reviewFinancialReport() reviewAttendanceSummary() monitorCommitteeActivities() superviseClubOperations()
6	Secretary	maintainCommitteeRecord() coordinateSchedule() reviewAttendanceSummary() monitorAdministrativeActivities()
7	Treasurer	receiveMembershipApplication() maintainMembershipRecord() recordMembershipPayment() recordDonation() recordIncome() recordExpense() generateFinancialReport() verifyGatewayPayment() identifyOverdueMemberships()

No.	Class	Identified methods
8	PublicitySecretary	createMediaPost() publishNotice() publishActivityMedia() updatePost() archivePost() setTargetAudience()
9	CulturalSecretary	createEvent() updateEvent() openEventRegistration() manageParticipants() coordinateEvent() closeEventRegistration()
10	FieldEquipmentSecretary	addEquipment() updateEquipment() checkAvailability() assignEquipment() recordReturn() createOperationalTask() assignTask() updateTaskProgress()
11	Activity	schedule() reschedule() cancel() addOrganizer() recordAttendance() recordManualAttendance() generateAttendanceSummary() listParticipants()
12	YogaSession	assignInstructor() openSession() closeSession() publishSessionSchedule()
13	Event	openRegistration() registerMember() removeRegistration() closeRegistration() publishParticipationInformation() listRegisteredMembers()
14	FinancialManagement	recordPayment() recordDonation() recordIncome() recordExpense() calculateBalance() reconcileTransaction() generateFinancialReport() getPaymentHistory() getDonationHistory() identifyOverdueMemberships()
15	PaymentGateway	processPayment() verifyTransaction() confirmPayment() getTransactionStatus() mapGatewayResponse()
16	MediaPost	create() update() publish() setAudience() attachMedia() archive() linkActivity()
17	YogaContent	create() update() addInstructions() setDifficulty() addRisk() addPrecaution() attachVideo() submitForApproval() approve() publish() search()

No.	Class	Identified methods
18	VajraAssistant	interpretQuestion() searchApprovedContent() generateAnswer() suggestRelatedContent() attachSafetyDisclaimer() clearConversationContext()
19	InventoryManagement	addEquipment() updateEquipment() removeEquipment() checkAvailability() assignEquipment() recordReturn() createTask() assignTask() updateTaskProgress() closeTask()

7. Finalizing Classes

The selected classes are finalized by checking responsibility placement, inheritance opportunities and unnecessary duplication. The following decisions produce the final high-level class model.

1. Authentication separation. Account credentials, status and permission behaviour are centralized in UserAccount. Member, YogaInstructor and ExecutiveCommitteeMember collaborate with it instead of duplicating authentication data.

2. Membership lifecycle. Applicant, MembershipRegistration and MembershipApplication are not retained as separate classes. Member represents both pending and active membership states. The static method Member.registerForMembership(...) creates a pending member and stores the submitted information.

3. Verification rule. The Treasurer receives the application and an ExecutiveCommitteeMember verifies its information. There is no separate formal approval operation. Membership becomes active only after verification and successful registration payment.

4. Committee hierarchy. ExecutiveCommitteeMember is a concrete superclass representing general executive members. President, Secretary, Treasurer, PublicitySecretary, CulturalSecretary and FieldEquipmentSecretary specialize it because they have distinct permissions and methods.

5. Activity hierarchy. Activity is abstract and contains shared schedule, organizer, participant and attendance behaviour. YogaSession and Event are its concrete subclasses.

6. Attendance placement. Attendance and AttendanceRecord are absorbed into Activity as attendanceEntries and attendance operations. A separate Attendance class would duplicate the activity context.

7. Event participation. EventParticipation and EventRegistration are absorbed into Event as registeredMembers, participationInformation and registration methods.

8. Financial consolidation. FinancialTransaction, payment, donation, income, expense and reports are maintained by FinancialManagement. PaymentGateway remains separate because it represents the external payment boundary.

9. Communication consolidation. Notice, announcement, activity post and event media share publication attributes and operations and are therefore merged into MediaPost.

10. Yoga knowledge consolidation. Pose information, instructions, risks, precautions and videos are consolidated in YogaContent. VajraAssistant remains separate because it provides conversational retrieval and safety guidance.

11. Inventory consolidation. Equipment, assignments, returns, operational tasks and helping-hand assignee information are maintained by InventoryManagement. Task and HelpingHand are not independent classes in this high-level model.

12. Presentation and automation. MemberDashboard, SystemAppearance and general SystemAutomation are boundary or presentation behaviours. They do not represent independent domain classes.

Final class count: 19.

UserAccount, Member, YogaInstructor, ExecutiveCommitteeMember, President, Secretary, Treasurer, PublicitySecretary, CulturalSecretary, FieldEquipmentSecretary, Activity, YogaSession, Event, FinancialManagement, PaymentGateway, MediaPost, YogaContent, VajraAssistant, InventoryManagement

8. Class Cards

Each class card records the selected attributes and methods, followed by the class responsibilities and principal collaborators. The format follows the reference document while retaining the detailed domain model developed above.

Table 10: Class Card of UserAccount

UserAccount	
Attributes accountId username passwordHash accountStatus accessPermissions lastLoginAt	Methods authenticate() login() logout() changePassword() activateAccount() deactivateAccount() grantPermission() revokePermission() updatePermissions()
Responsibilities	**Collaborative Classes**
Authenticate users Maintain account status Enforce role-based access	Member YogaInstructor ExecutiveCommitteeMember

Table 11: Class Card of Member

Member	
Attributes memberId fullName contactInformation photograph dateOfBirth applicationStatus verificationStatus registrationPaymentStatus membershipNumber membershipStatus qrCode cardInformation registrationDate membershipStartDate account	Methods static registerForMembership() submitApplication() updateApplication() viewApplicationStatus() markInformationVerified() requestApplicationCorrection() recordRegistrationPayment() activateMembership() generateMembershipNumber() generateQRCode() issueMembershipCard() updateProfile() payFee() donate() registerForEvent() viewPaymentHistory() viewDonationHistory() viewUpcomingActivities() viewYogaContent()
Responsibilities	**Collaborative Classes**
Register for membership Maintain member identity and status Make payments and donations Participate in events Access member information	UserAccount ExecutiveCommitteeMember Activity Event FinancialManagement YogaContent

Table 12: Class Card of YogaInstructor

YogaInstructor	
Attributes instructorId fullName contactInformation expertise account authoredContent	Methods createYogaContent() updateYogaContent() addPoseInstructions() setDifficultyLevel() addRiskAndPrecaution() attachDemonstrationVideo() submitContentForApproval() viewAssignedSessions()
Responsibilities	**Collaborative Classes**
Provide yoga instruction Author safe yoga content Maintain professional information	UserAccount YogaSession YogaContent

Table 13: Class Card of ExecutiveCommitteeMember

ExecutiveCommitteeMember	
Attributes committeeMemberId member termStartDate termEndDate committeeStatus account assignedPermissions	Methods verifyMemberInformation() requestMemberCorrection() recordAttendanceByQR() recordAttendanceManually() generateAttendanceSummary() viewMemberDirectory() manageCommitteeTerm() createActivity() updateActivity()
Responsibilities	**Collaborative Classes**
Verify applicant information Perform common committee administration Record attendance Maintain committee term and access	UserAccount Member Activity FinancialManagement MediaPost InventoryManagement

Table 14: Class Card of President

President	
Attributes inherits ExecutiveCommitteeMember attributes oversightNotes	Methods reviewFinancialReport() reviewAttendanceSummary() monitorCommitteeActivities() superviseClubOperations()
Responsibilities	**Collaborative Classes**
Supervise club operation Review organizational and financial summaries	ExecutiveCommitteeMember FinancialManagement Activity Secretary

Table 15: Class Card of Secretary

Secretary	
Attributes inherits ExecutiveCommitteeMember attributes committeeRecords coordinatedSchedules	Methods maintainCommitteeRecord() coordinateSchedule() reviewAttendanceSummary() monitorAdministrativeActivities()
Responsibilities	**Collaborative Classes**
Maintain committee records Coordinate schedules Support organizational supervision	ExecutiveCommitteeMember President Activity

Table 16: Class Card of Treasurer

Treasurer	
Attributes inherits ExecutiveCommitteeMember attributes receivedApplications managedFinancialRecords	Methods receiveMembershipApplication() maintainMembershipRecord() recordMembershipPayment() recordDonation() recordIncome() recordExpense() generateFinancialReport() verifyGatewayPayment() identifyOverdueMemberships()
Responsibilities	**Collaborative Classes**
Receive membership applications Maintain membership and financial records Submit financial reports	ExecutiveCommitteeMember Member FinancialManagement PaymentGateway President Secretary

Table 17: Class Card of PublicitySecretary

PublicitySecretary	
Attributes inherits ExecutiveCommitteeMember attributes managedPosts	Methods createMediaPost() publishNotice() publishActivityMedia() updatePost() archivePost() setTargetAudience()
Responsibilities	**Collaborative Classes**
Publish official communication Manage activity media and visibility	ExecutiveCommitteeMember MediaPost Activity Event

Table 18: Class Card of CulturalSecretary

CulturalSecretary	
Attributes inherits ExecutiveCommitteeMember attributes managedEvents	Methods createEvent() updateEvent() openEventRegistration() manageParticipants() coordinateEvent() closeEventRegistration()
Responsibilities	**Collaborative Classes**
Create cultural and special events Manage registration and participants Coordinate event operations	ExecutiveCommitteeMember Event Member FinancialManagement MediaPost

Table 19: Class Card of FieldEquipmentSecretary

FieldEquipmentSecretary	
Attributes inherits ExecutiveCommitteeMember attributes managedEquipment managedTasks	Methods addEquipment() updateEquipment() checkAvailability() assignEquipment() recordReturn() createOperationalTask() assignTask() updateTaskProgress()
Responsibilities	**Collaborative Classes**
Maintain equipment Assign equipment to activities Coordinate operational tasks	ExecutiveCommitteeMember InventoryManagement Activity

Table 20: Class Card of Activity

Activity	
Attributes activityId title activityDate startTime location status organizers participants attendanceEntries	Methods schedule() reschedule() cancel() addOrganizer() recordAttendance() recordManualAttendance() generateAttendanceSummary() listParticipants()
Responsibilities	**Collaborative Classes**
Provide common activity schedule Maintain attendance and participants Support activity lifecycle	Member ExecutiveCommitteeMember MediaPost FinancialManagement InventoryManagement

Table 21: Class Card of YogaSession

YogaSession	
Attributes inherits Activity attributes instructor duration regularSession	Methods assignInstructor() openSession() closeSession() publishSessionSchedule()
Responsibilities	**Collaborative Classes**
Represent a scheduled yoga session Connect instructor and session schedule	Activity YogaInstructor Member

Table 22: Class Card of Event

Event	
Attributes inherits Activity attributes eventType description instructions registrationRequired registrationOpen registeredMembers participationInformation	Methods openRegistration() registerMember() removeRegistration() closeRegistration() publishParticipationInformation() listRegisteredMembers()
Responsibilities	**Collaborative Classes**
Represent cultural or special events Manage event registration Maintain participant information	Activity Member CulturalSecretary MediaPost FinancialManagement

Table 23: Class Card of FinancialManagement

FinancialManagement	
Attributes membershipPayments donations incomeEntries expenseEntries currentBalance generatedReports	Methods recordPayment() recordDonation() recordIncome() recordExpense() calculateBalance() reconcileTransaction() generateFinancialReport() getPaymentHistory() getDonationHistory() identifyOverdueMemberships()
Responsibilities	**Collaborative Classes**
Maintain financial entries Calculate totals and balance Generate monthly and yearly reports	Member Treasurer President Secretary PaymentGateway Event

Table 24: Class Card of PaymentGateway

PaymentGateway	
Attributes gatewayName supportedMethods availability transactionReferences lastGatewayResponse	Methods processPayment() verifyTransaction() confirmPayment() getTransactionStatus() mapGatewayResponse()
Responsibilities	**Collaborative Classes**
Connect to external payment facility Verify and confirm online payments	FinancialManagement Treasurer Member

Table 25: Class Card of MediaPost

MediaPost	
Attributes postId postType title body attachments visibility targetAudience publishDate relatedActivity archiveStatus	Methods create() update() publish() setAudience() attachMedia() archive() linkActivity()
Responsibilities	**Collaborative Classes**
Represent notices and activity media Control publication and audience Preserve previous posts	PublicitySecretary Activity Event Member

Table 26: Class Card of YogaContent

YogaContent	
Attributes contentId contentType title instructions difficultyLevel risks precautions videoReference visibility approvalStatus author createdAt updatedAt	Methods create() update() addInstructions() setDifficulty() addRisk() addPrecaution() attachVideo() submitForApproval() approve() publish() search()
Responsibilities	**Collaborative Classes**
Store approved yoga knowledge Maintain instructions and safety information Provide searchable public or member content	YogaInstructor VajraAssistant Member

Table 27: Class Card of VajraAssistant

VajraAssistant	
Attributes approvedContentSource currentQuestion interpretedTopic generatedAnswer suggestions safetyDisclaimer	Methods interpretQuestion() searchApprovedContent() generateAnswer() suggestRelatedContent() attachSafetyDisclaimer() clearConversationContext()
Responsibilities	**Collaborative Classes**
Answer yoga questions from approved content Recommend learning material Present safety guidance without diagnosis	YogaContent Member YogaInstructor

Table 28: Class Card of InventoryManagement

InventoryManagement	
Attributes equipmentCatalogue availabilityRecords equipmentAssignments returnRecords operationalTasks taskAssignees progressRecords	Methods addEquipment() updateEquipment() removeEquipment() checkAvailability() assignEquipment() recordReturn() createTask() assignTask() updateTaskProgress() closeTask()
Responsibilities	**Collaborative Classes**
Maintain inventory and availability Assign equipment to activities Maintain operational tasks and assignees	FieldEquipmentSecretary Activity FinancialManagement

9. CRC Diagram

Each figure is centered on one class and shows every collaborator identified in its class card.

Ref: Table 10

Name: UserAccount

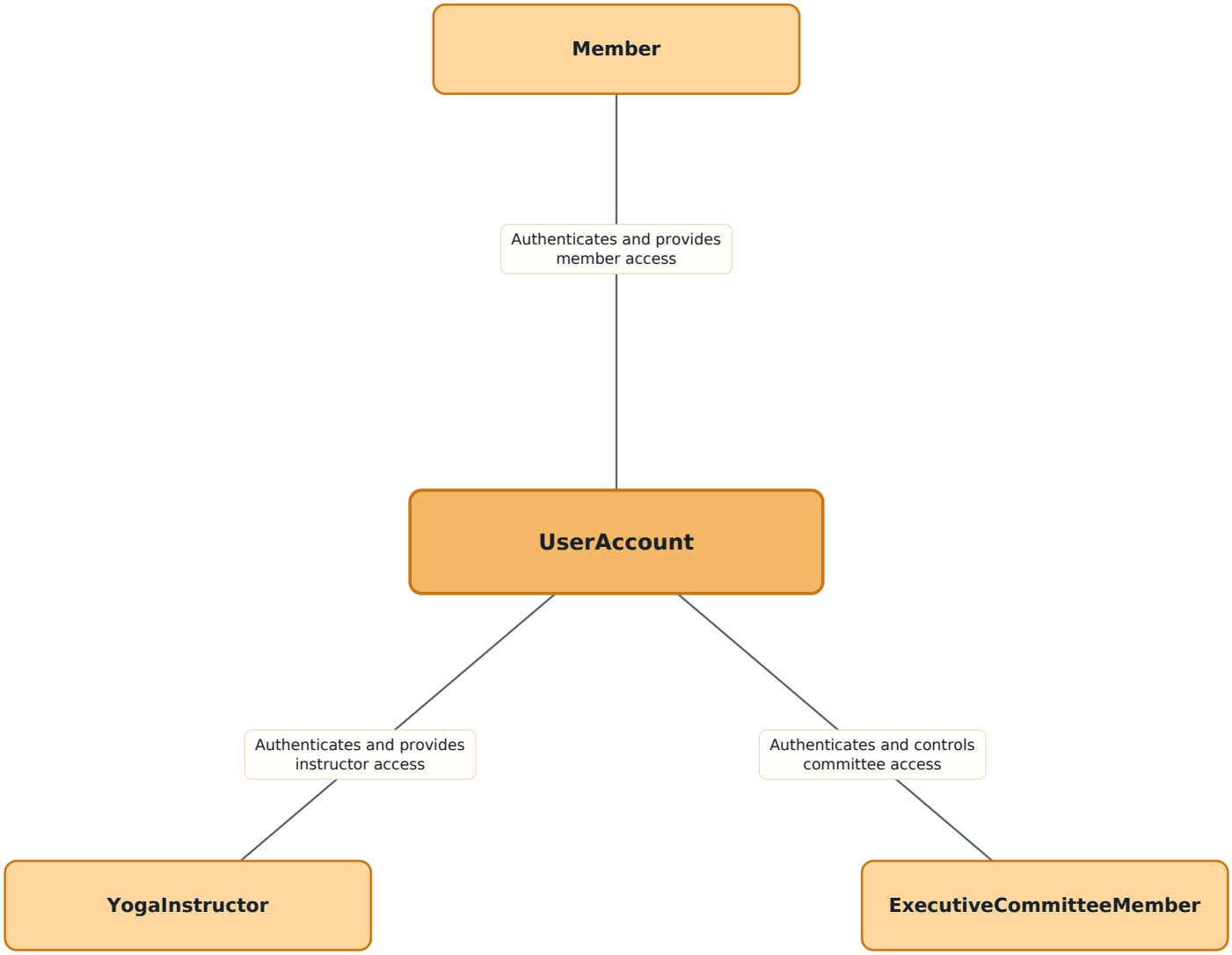


Fig crc.1: UserAccount

Ref: Table 11

Name: Member

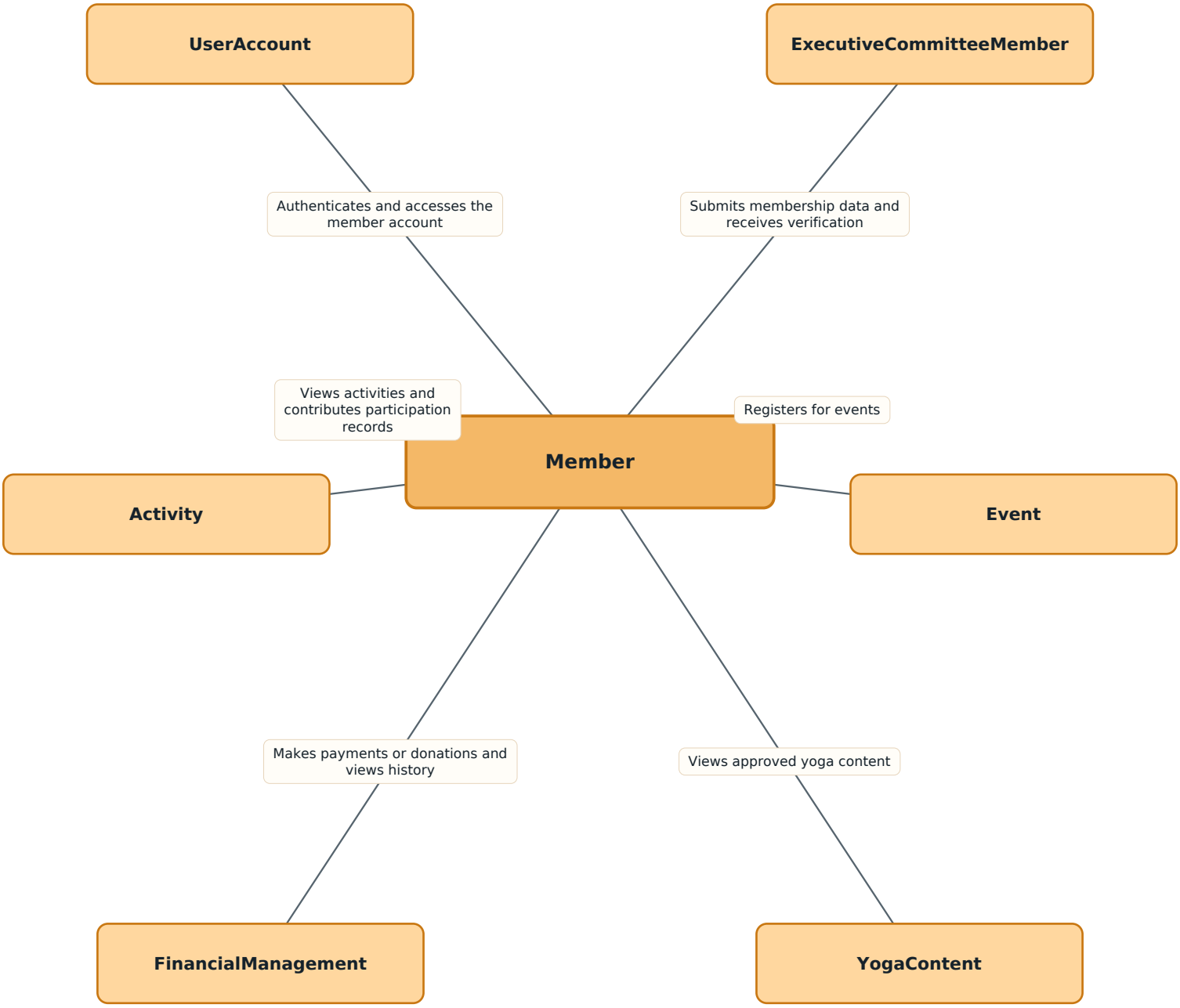


Fig crc.2: Member

Ref: Table 12

Name: YogaInstructor

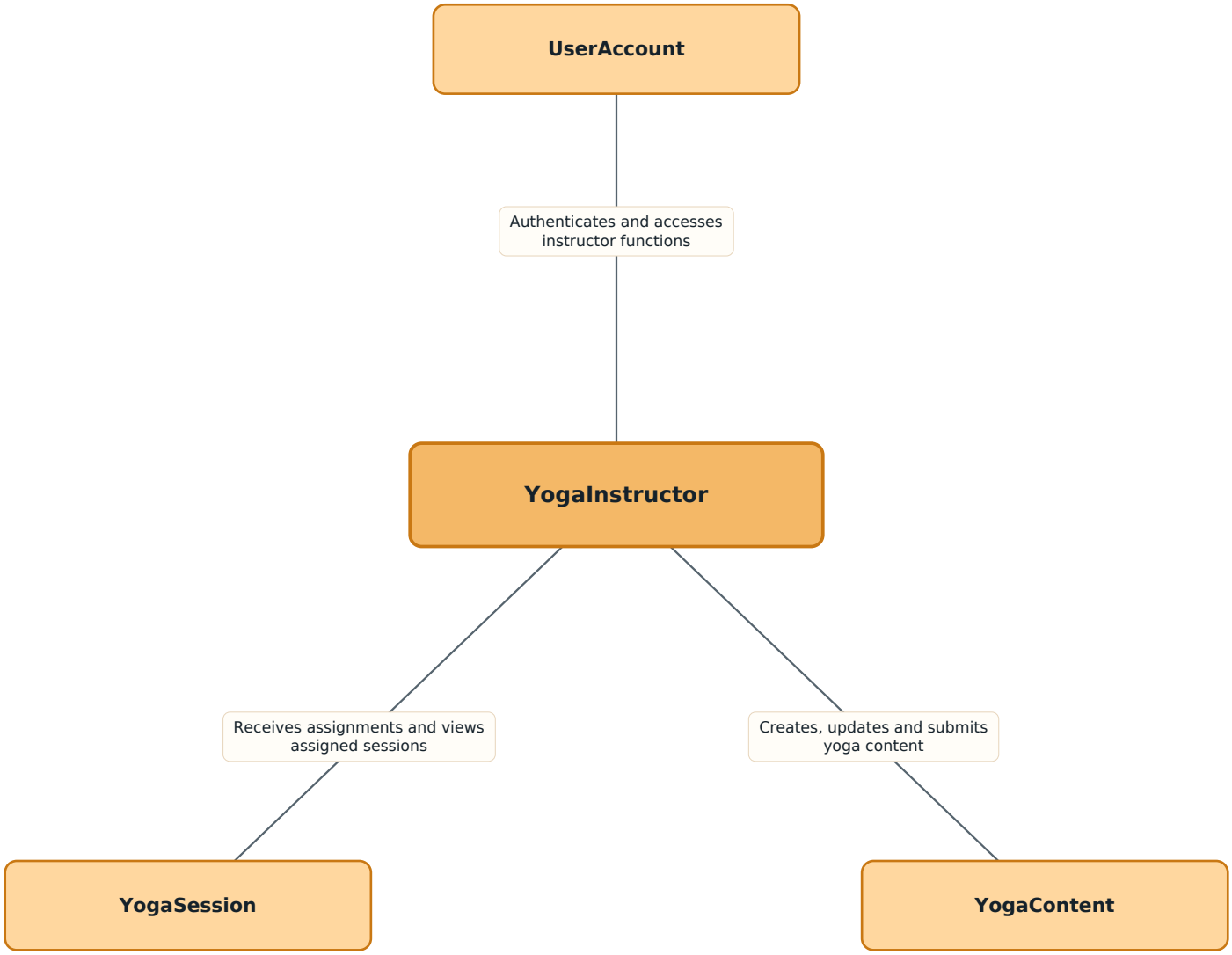


Fig crc.3: YogaInstructor

Ref: Table 13

Name: ExecutiveCommitteeMember

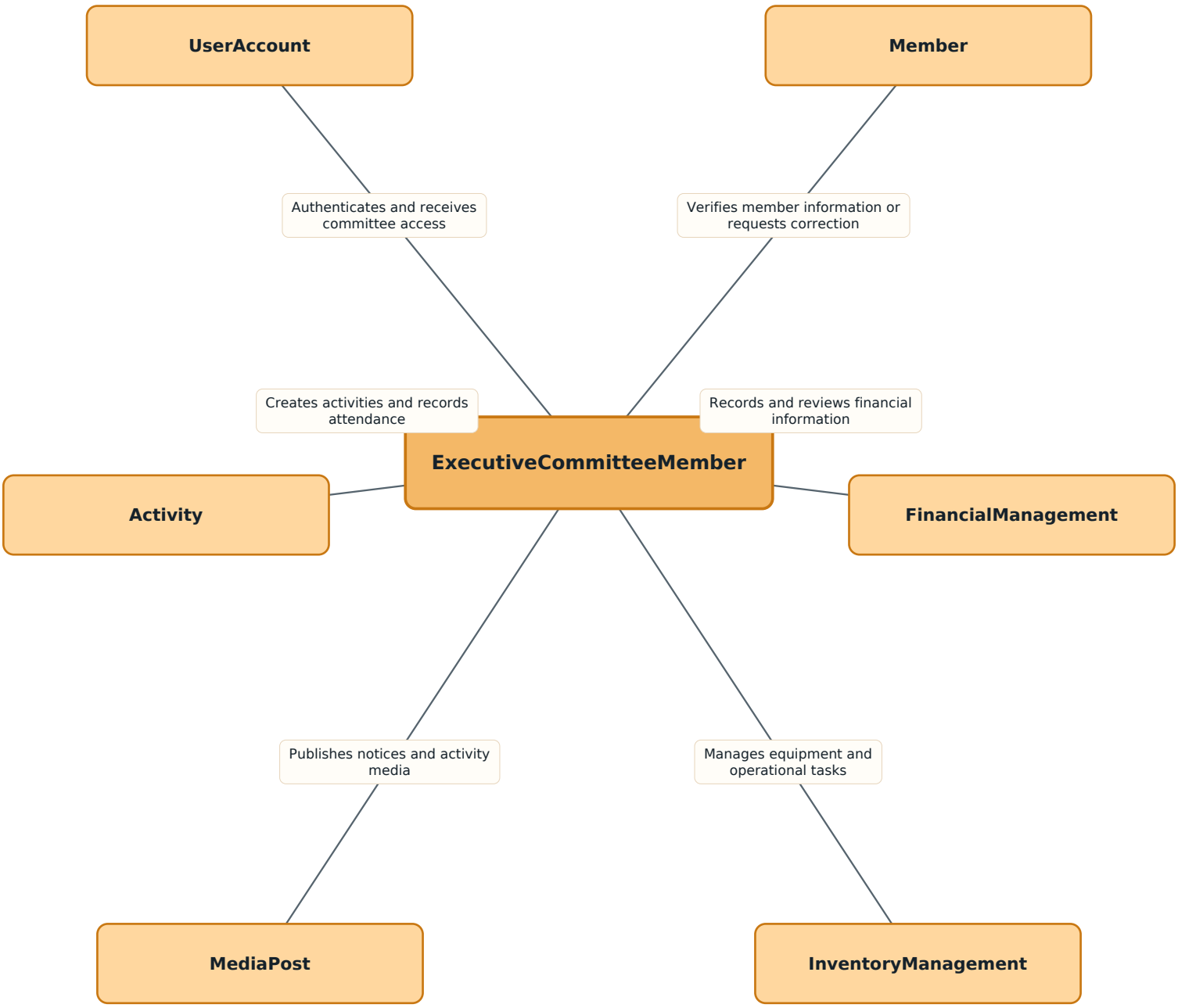


Fig crc.4: ExecutiveCommitteeMember

Ref: Table 14

Name: President

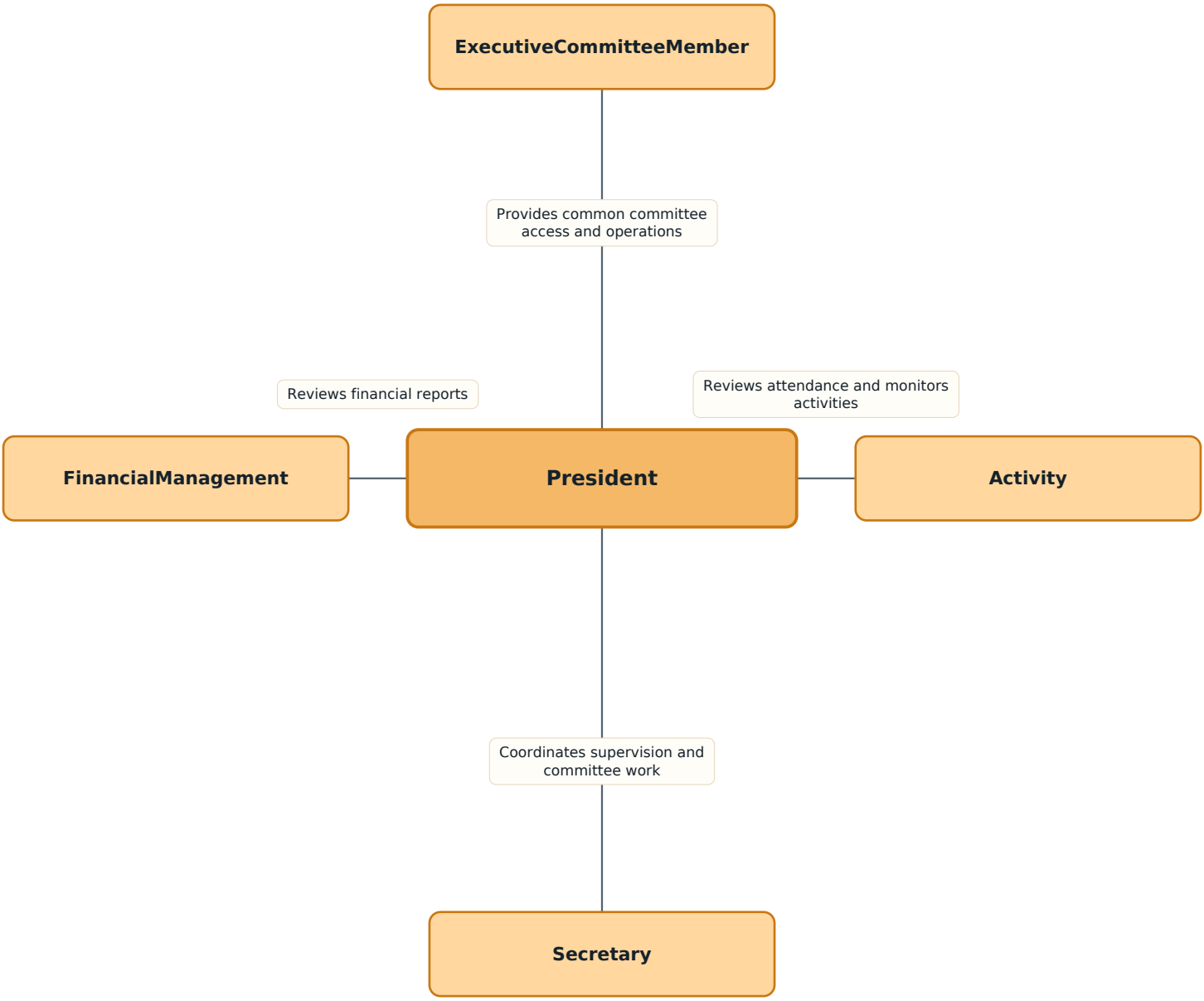


Fig crc.5: President

Ref: Table 15

Name: Secretary

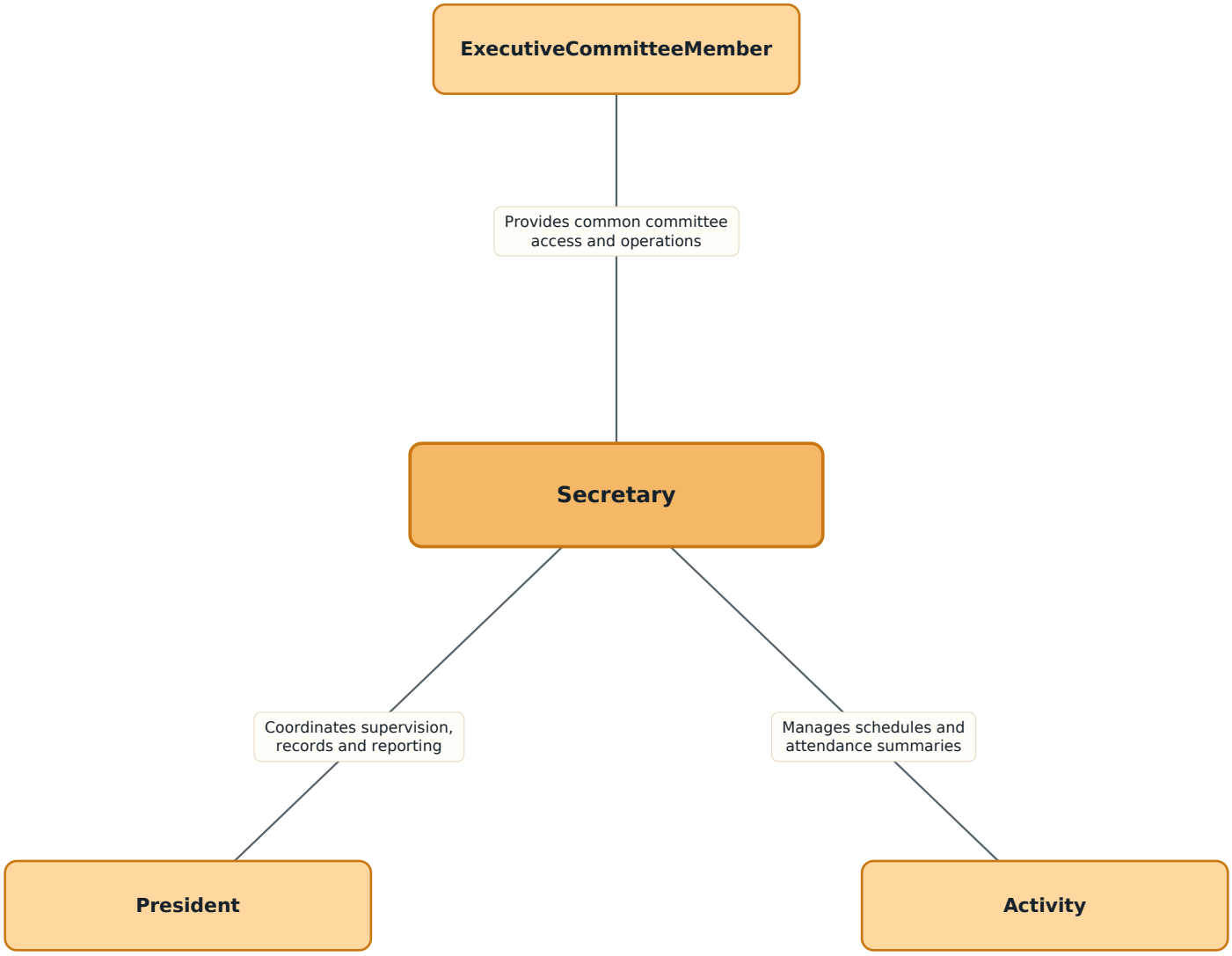


Fig crc.6: Secretary

Ref: Table 16

Name: Treasurer

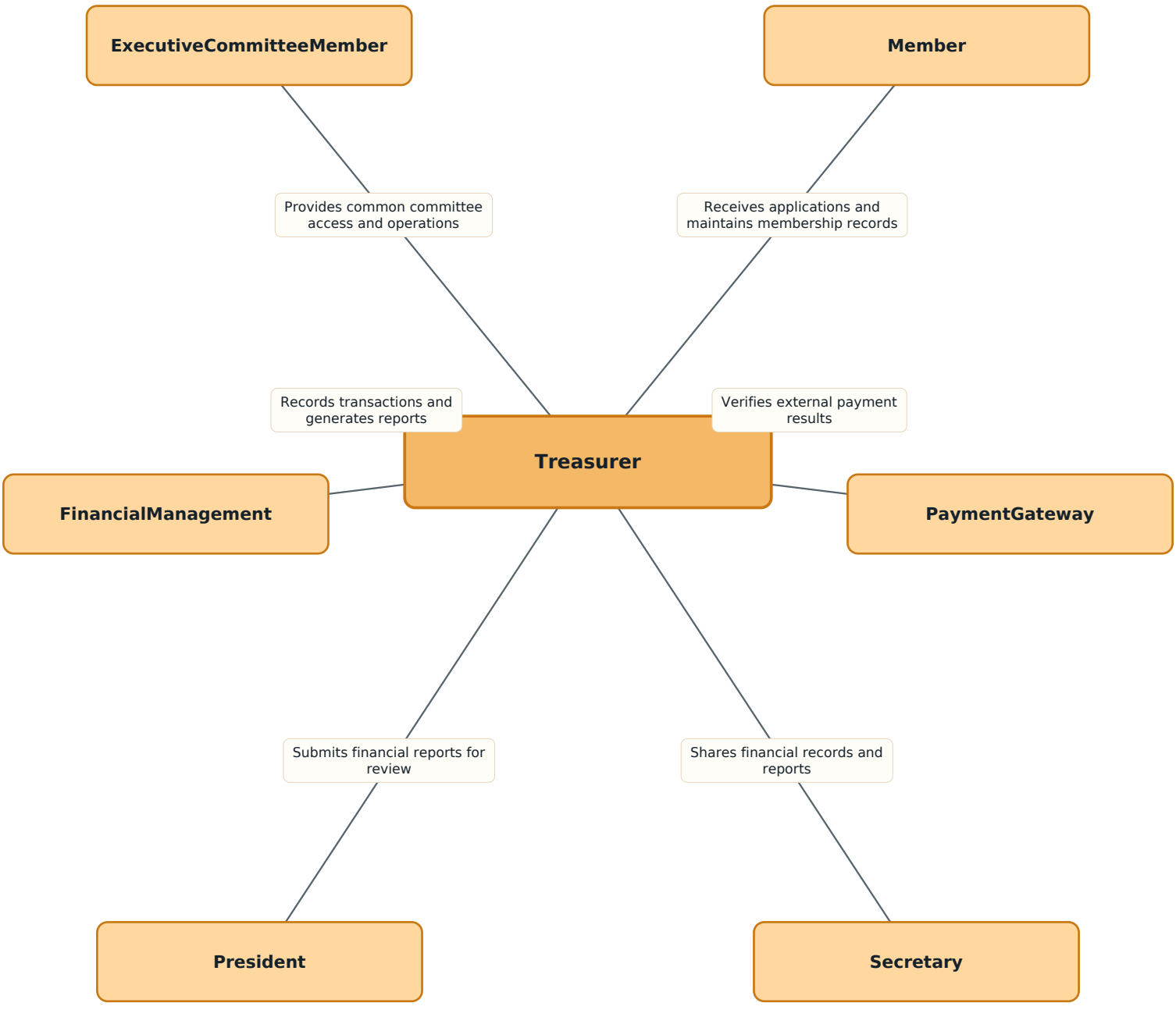


Fig crc.7: Treasurer

Ref: Table 17

Name: PublicitySecretary

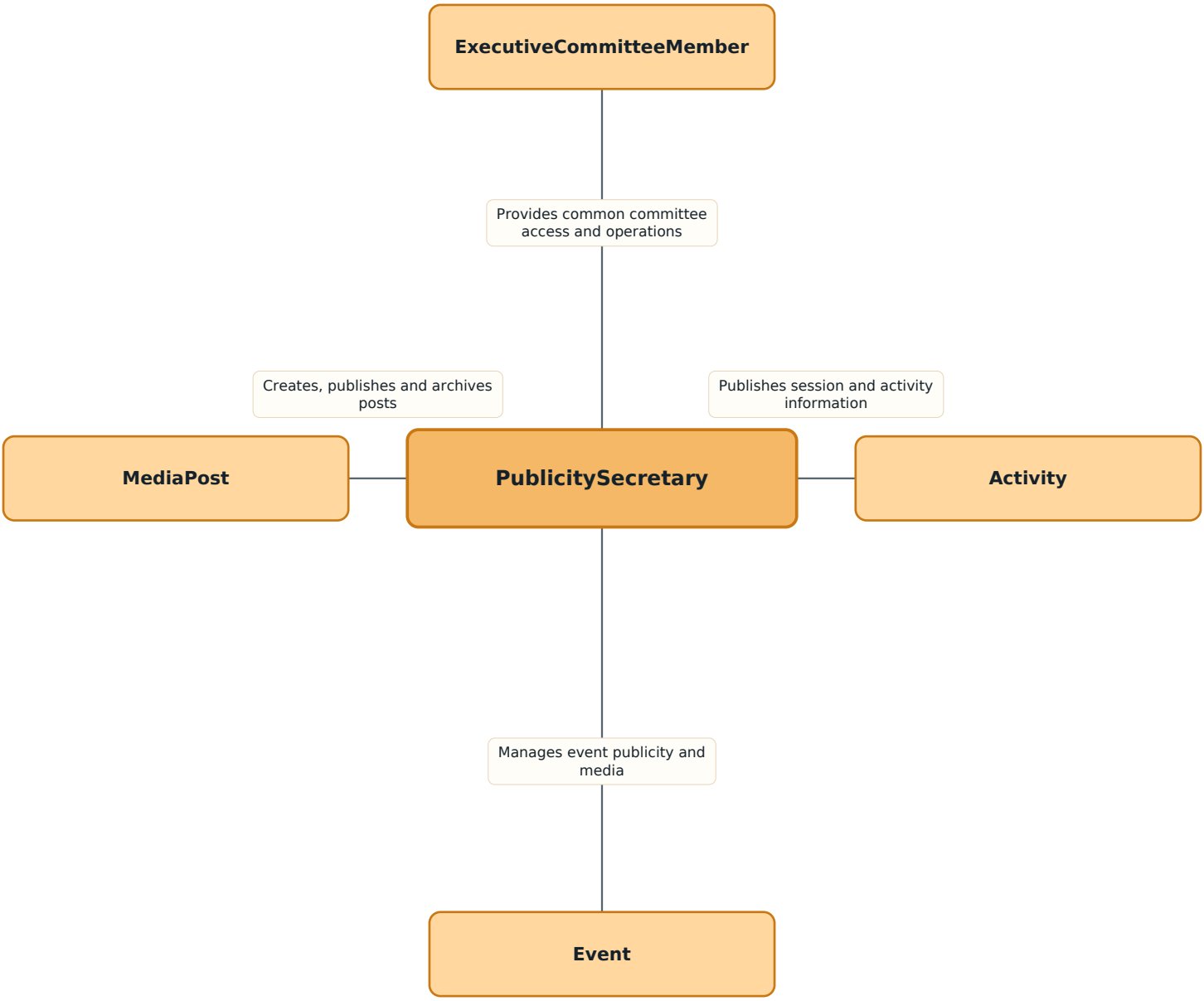


Fig crc.8: PublicitySecretary

Ref: Table 18

Name: CulturalSecretary

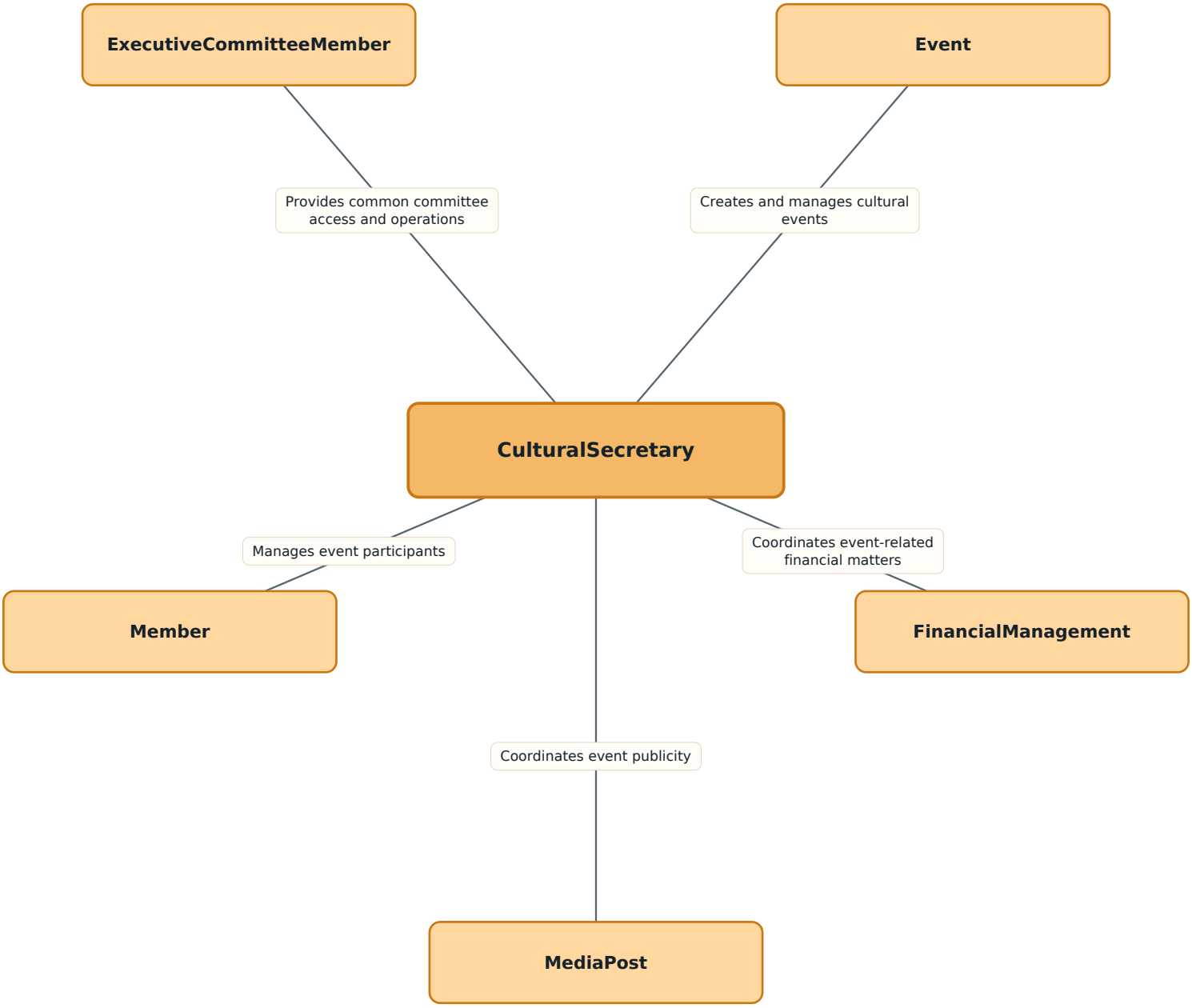


Fig crc.9: CulturalSecretary

Ref: Table 19

Name: FieldEquipmentSecretary

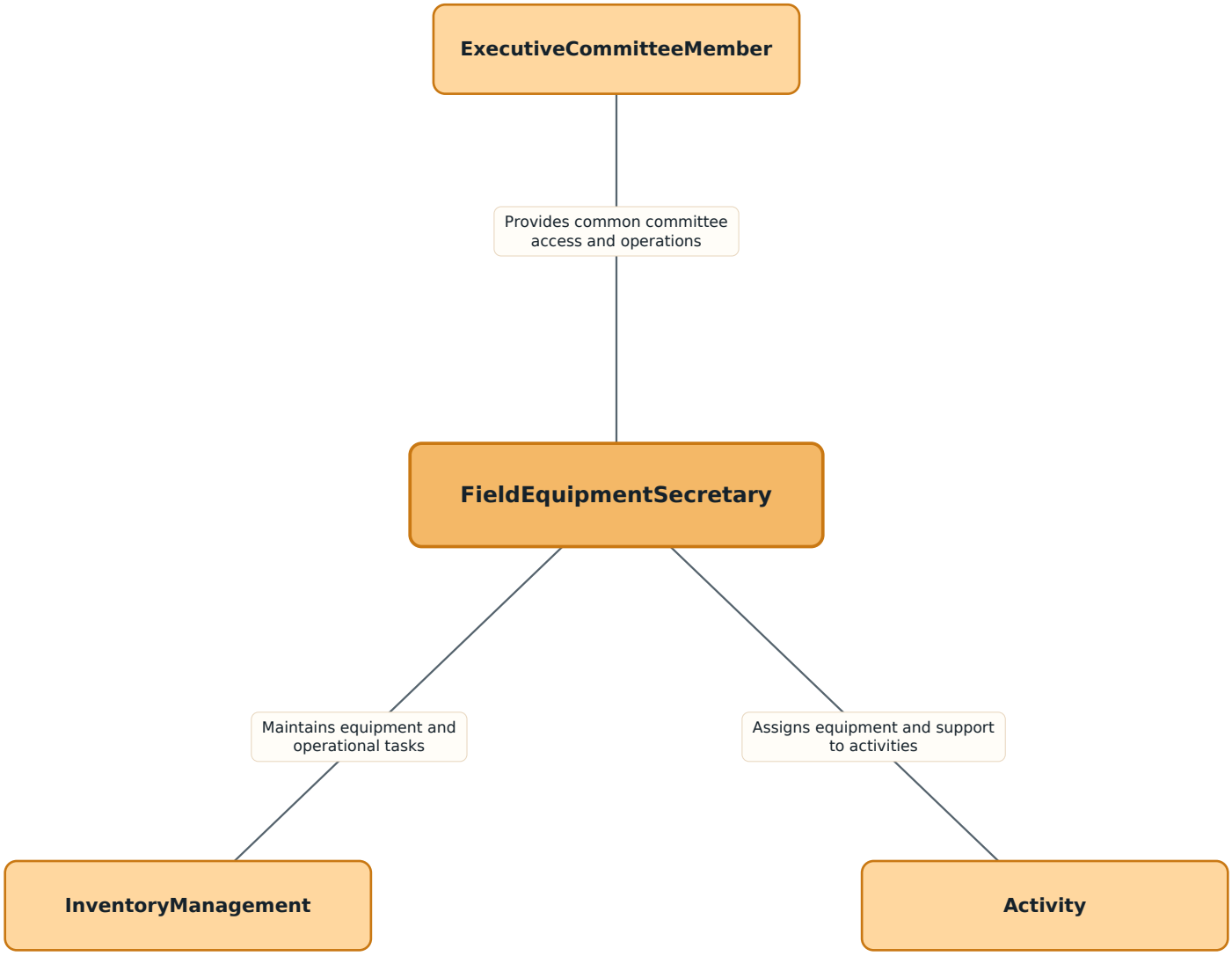


Fig crc.10: FieldEquipmentSecretary

Ref: Table 20

Name: Activity

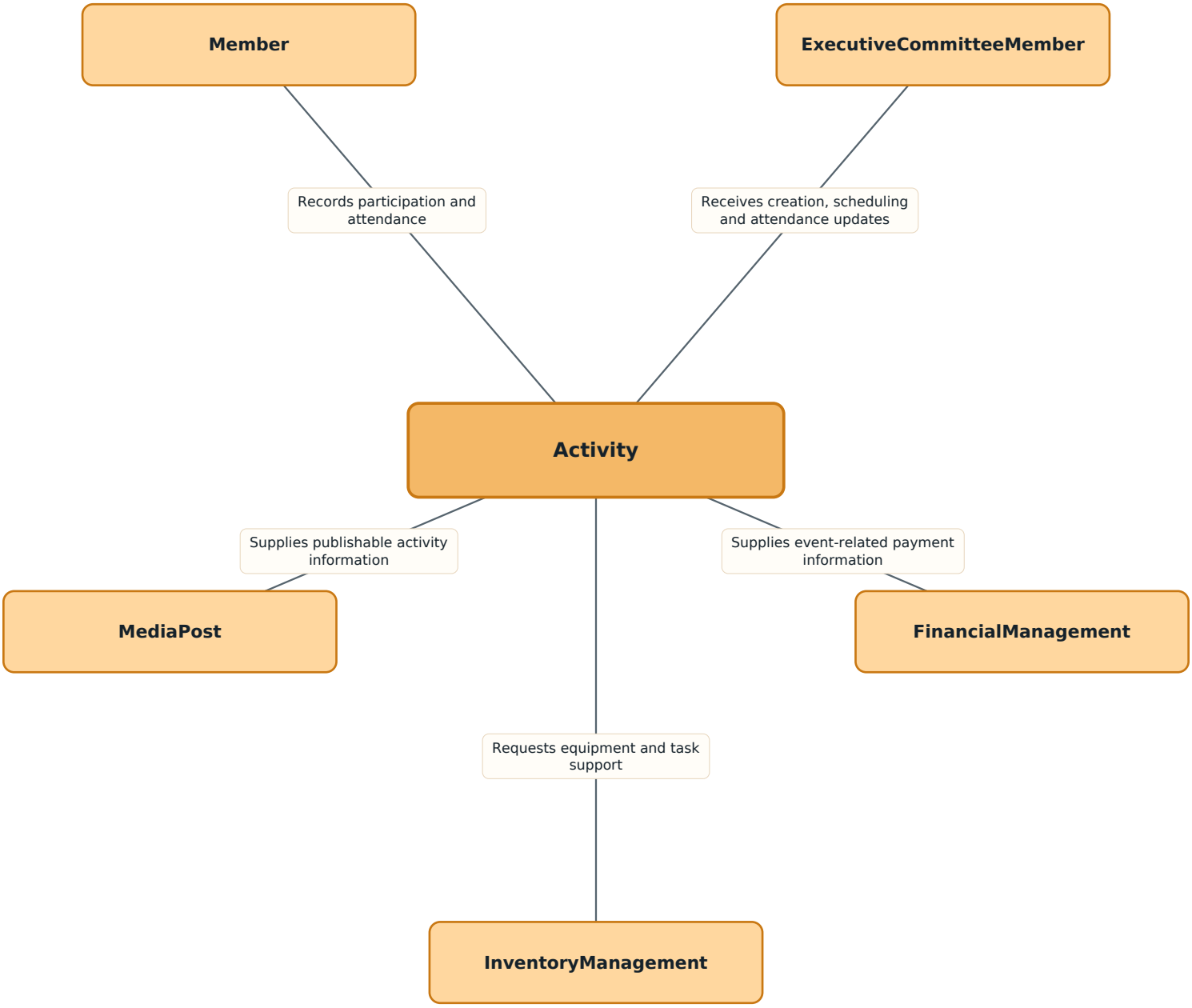


Fig crc.11: Activity

Ref: Table 21

Name: YogaSession

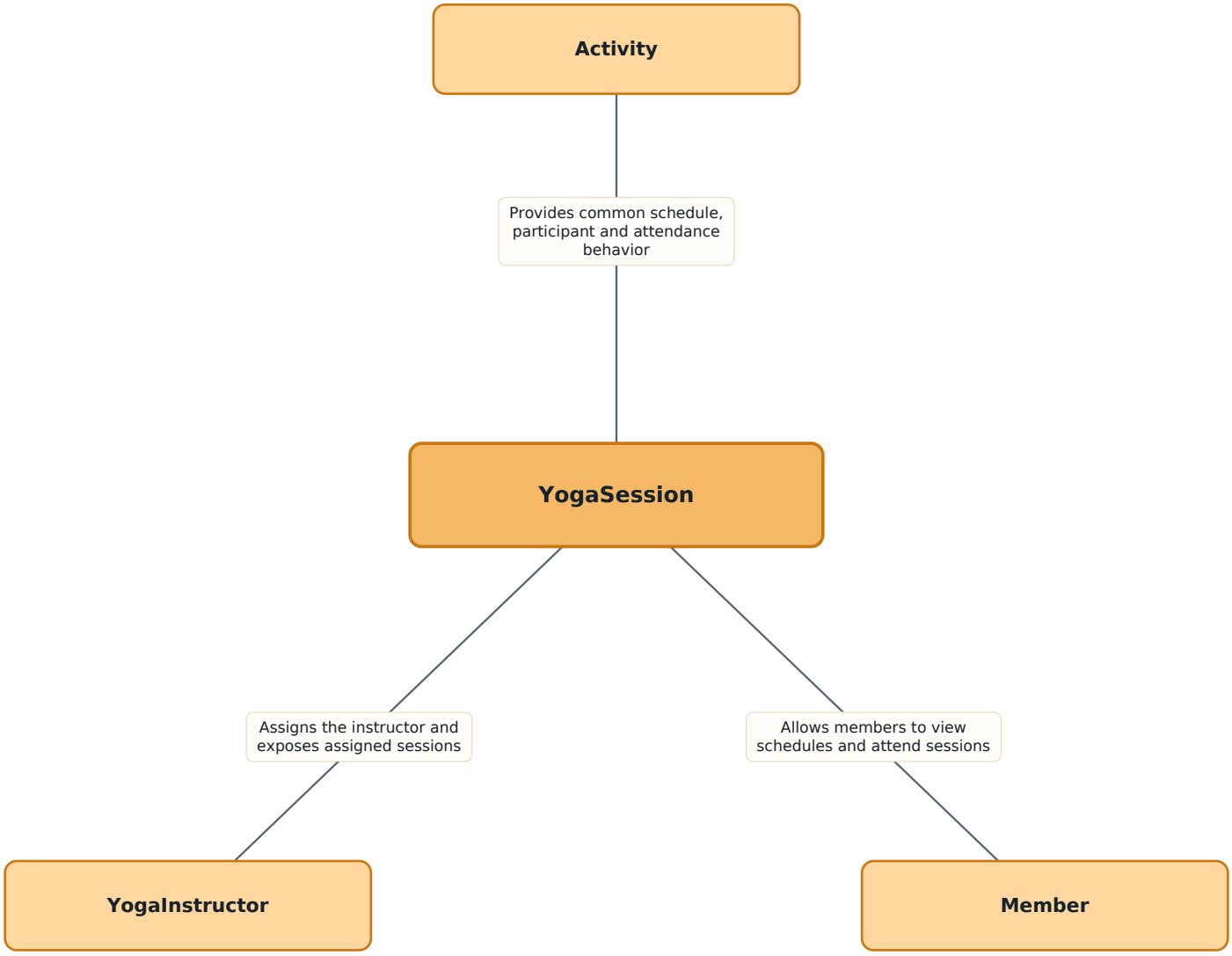


Fig crc.12: YogaSession

Ref: Table 22

Name: Event

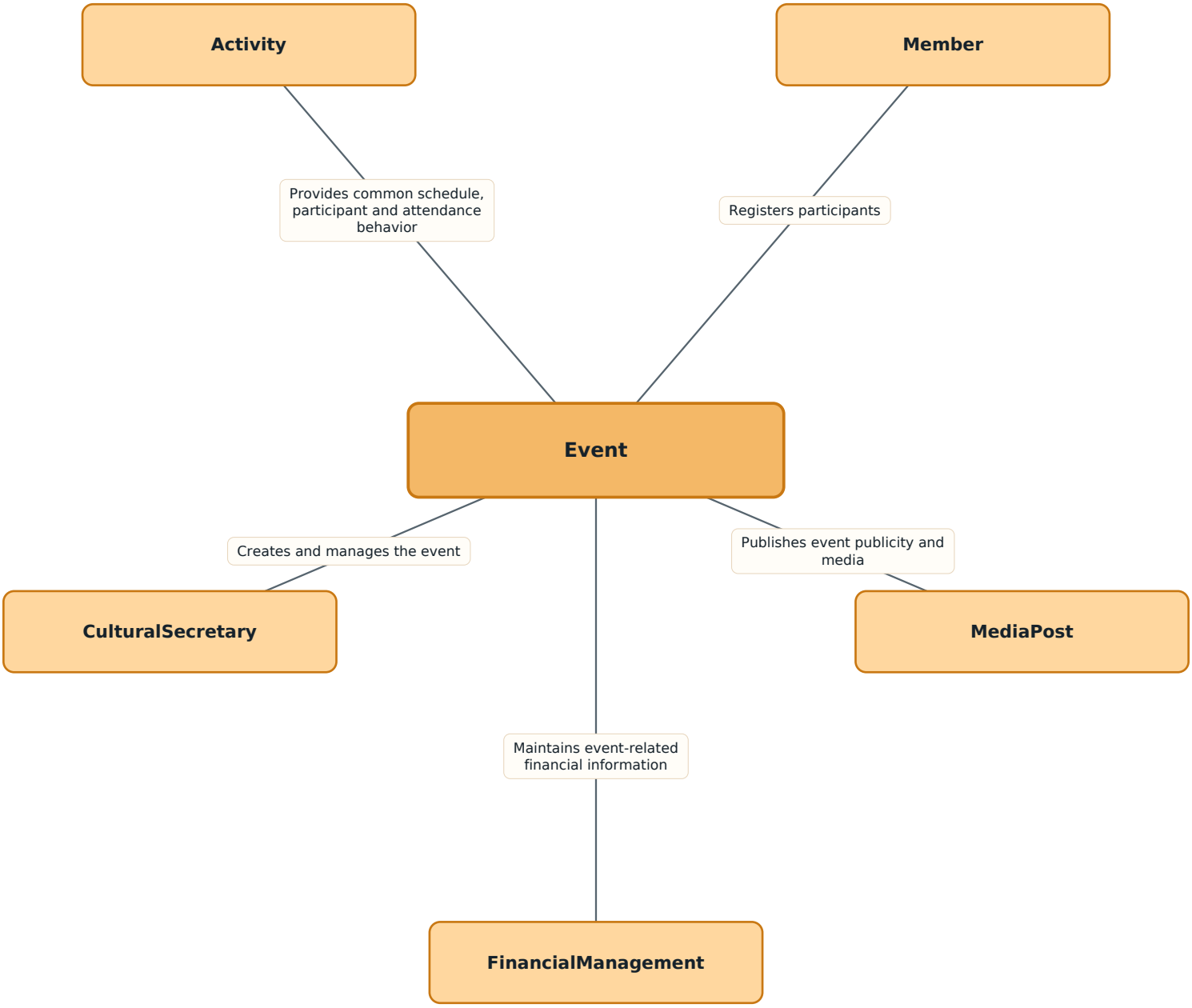


Fig crc.13: Event

Ref: Table 23

Name: FinancialManagement

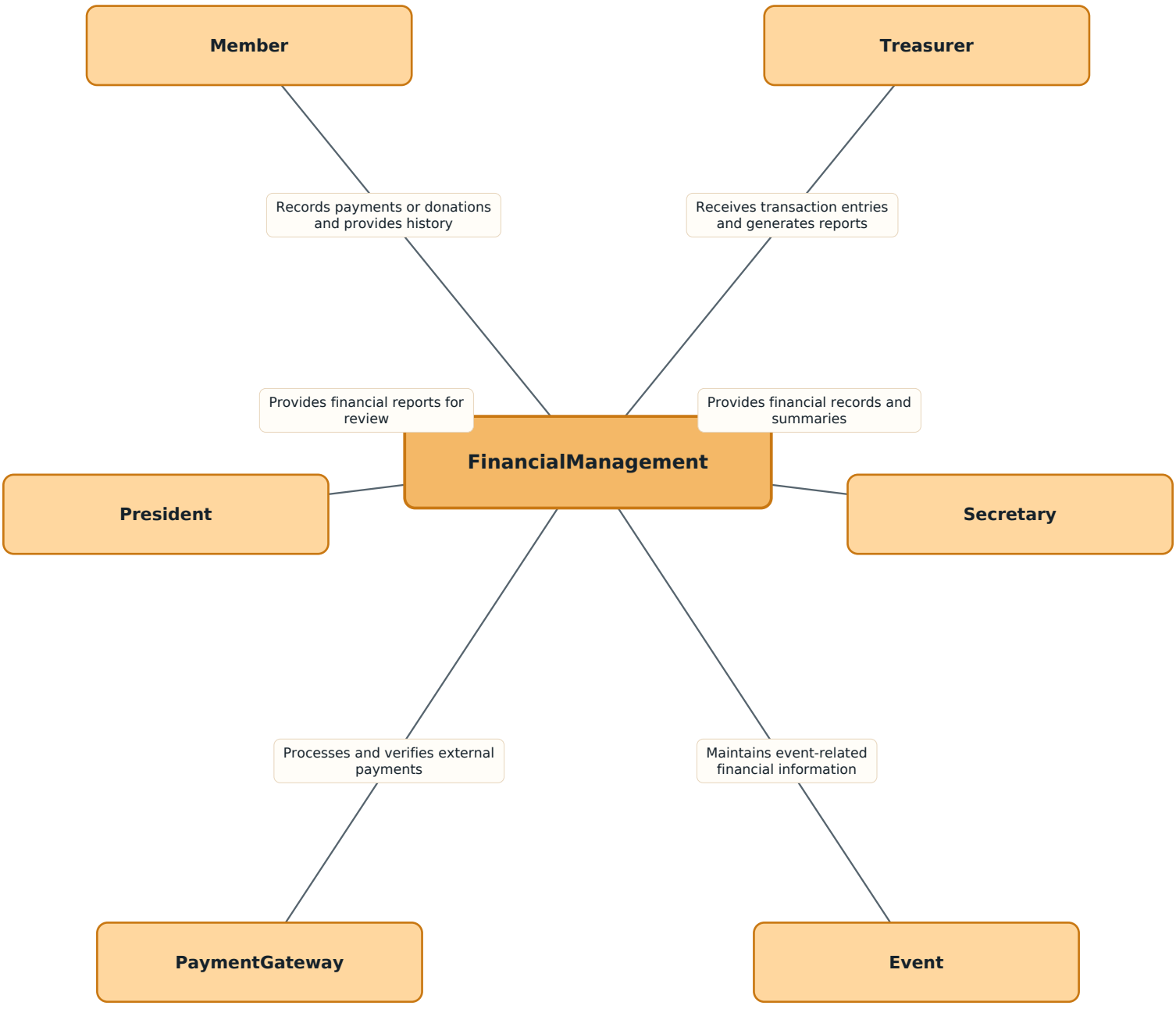


Fig crc.14: FinancialManagement

Ref: Table 24

Name: PaymentGateway

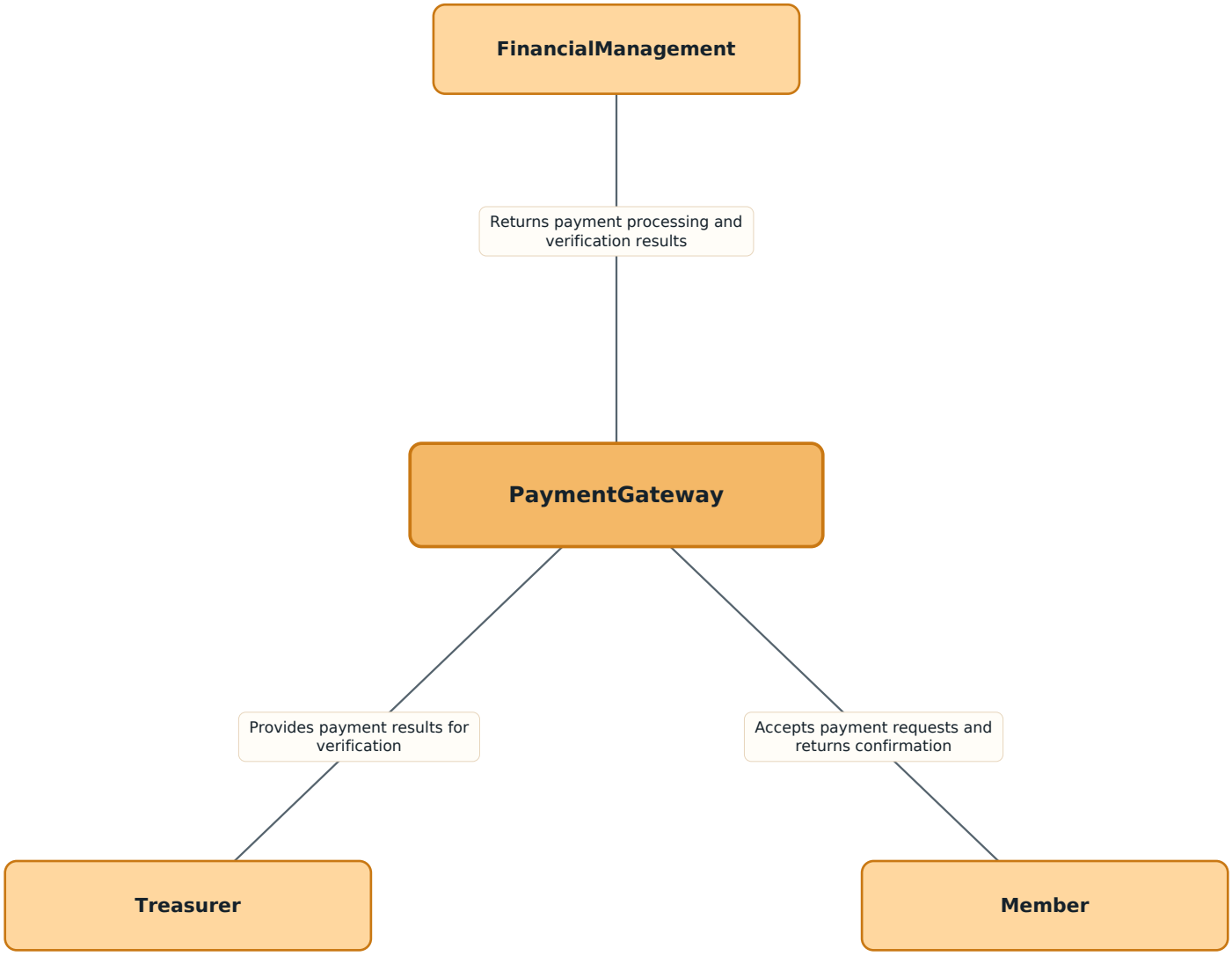


Fig crc.15: PaymentGateway

Ref: Table 25

Name: MediaPost

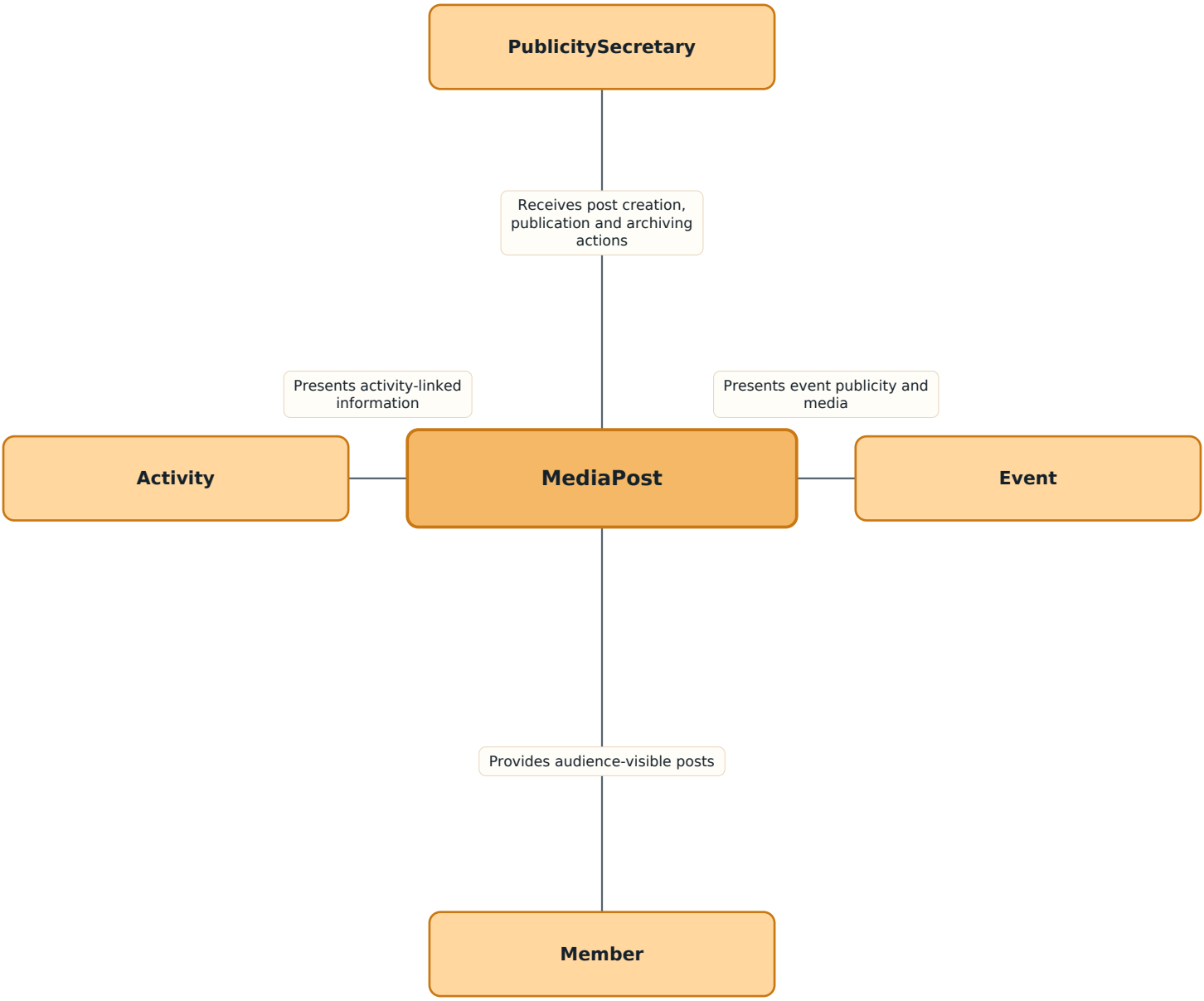


Fig crc.16: MediaPost

Ref: Table 26

Name: YogaContent

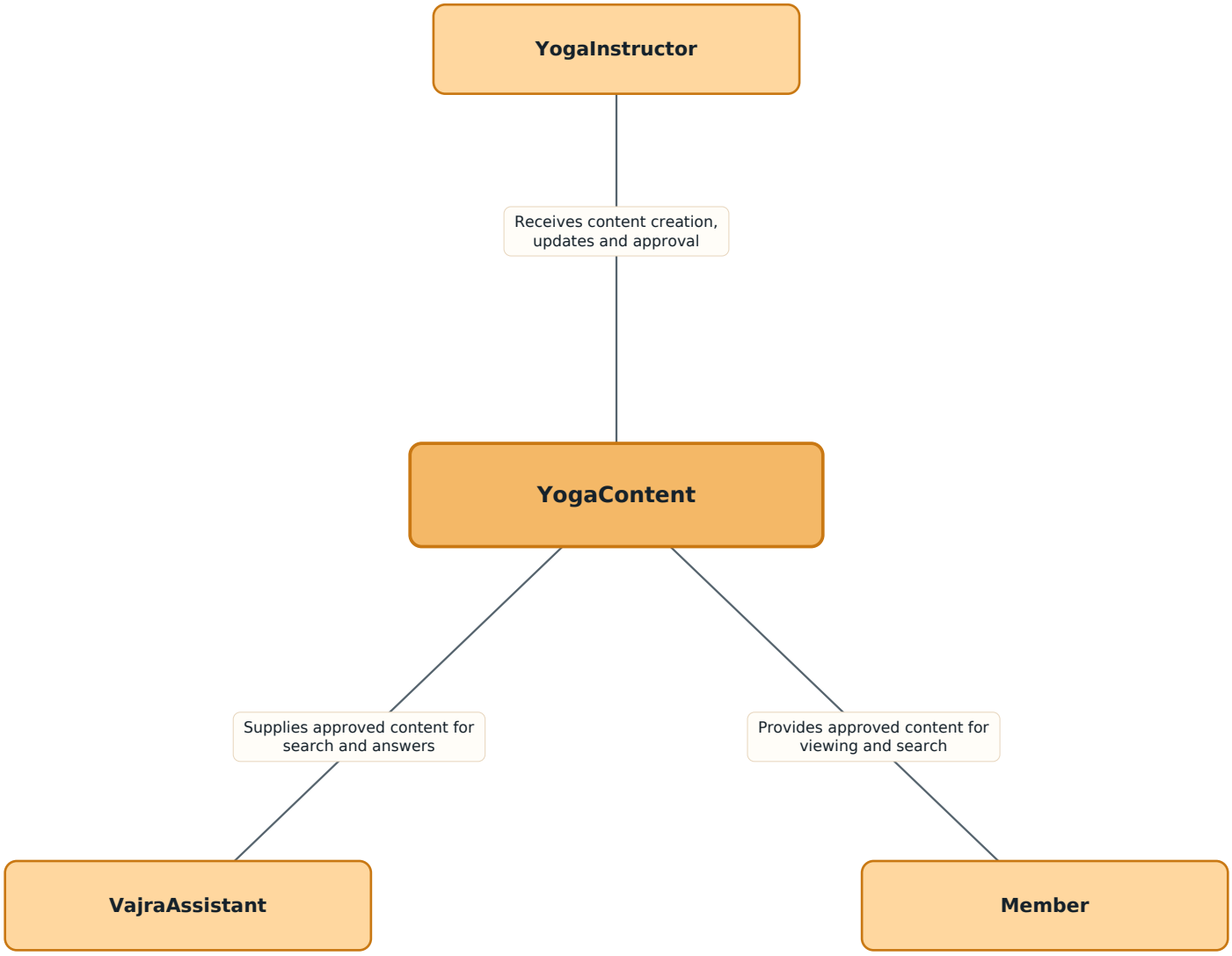


Fig crc.17: YogaContent

Ref: Table 27

Name: VajraAssistant

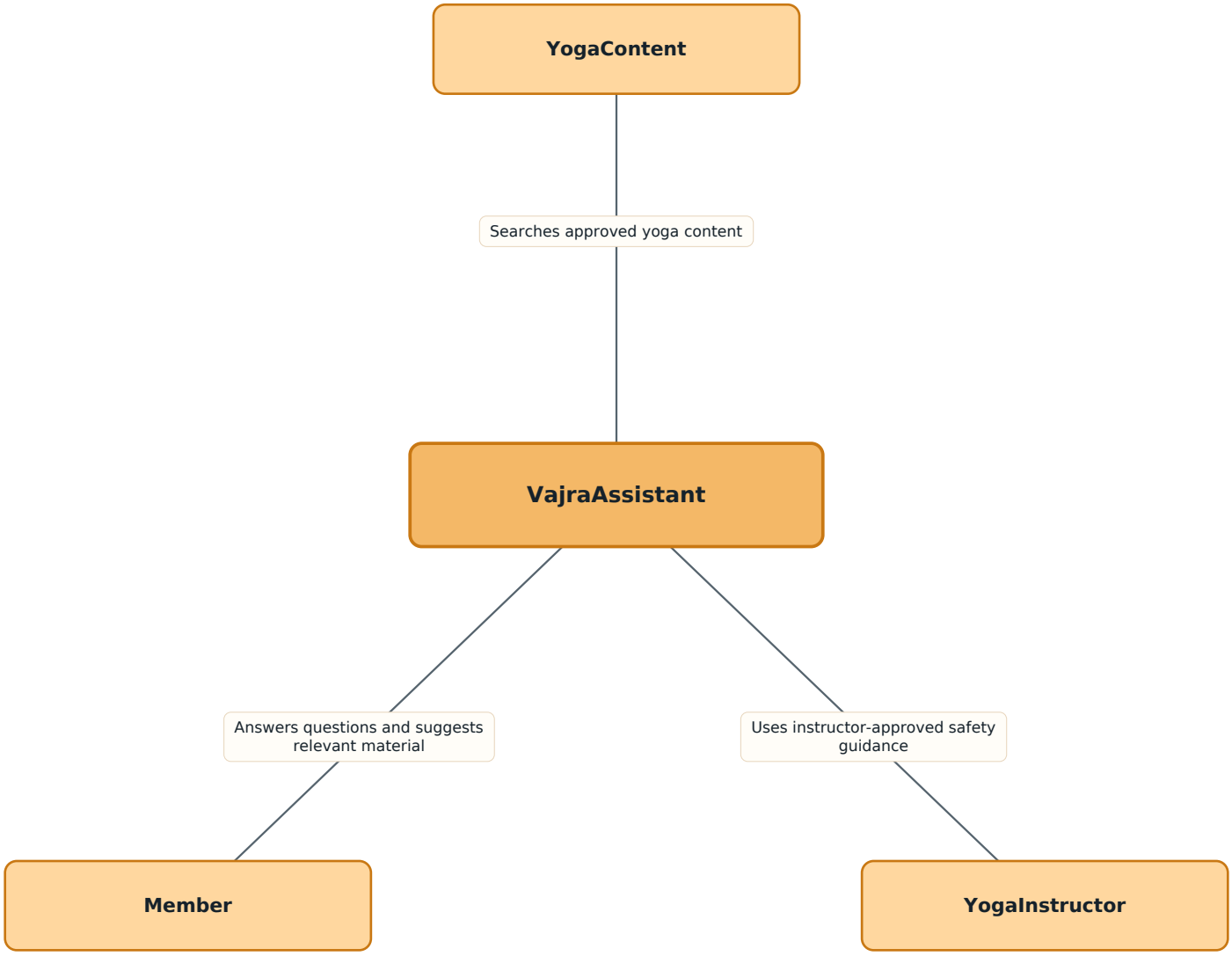


Fig crc.18: VajraAssistant

Ref: Table 28

Name: InventoryManagement

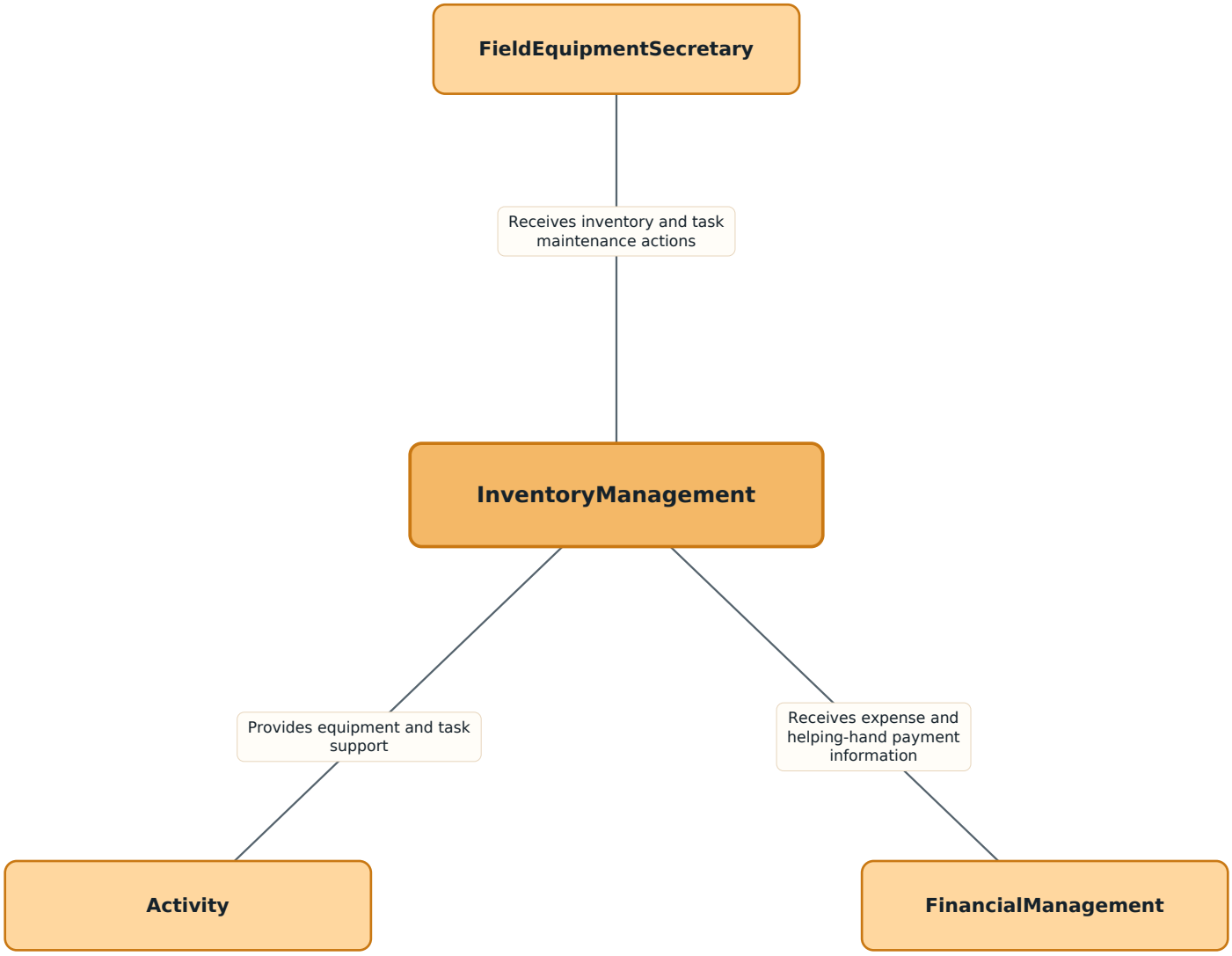


Fig crc.19: InventoryManagement